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ELEC 2100 Signals and Systems

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Outline Chapter 5 Laplace Transform



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1 Overview

2 Definition of Laplace Transform

3 Laplace Transform of Common Signals

4 Basic Properties of Laplace Transform

5 Inverse Laplace Transform

6 LT Analysis of Dynamic Circuits

7 System Functions

8 Relationship Between LT and FT

1. Introduction of Laplace Transform

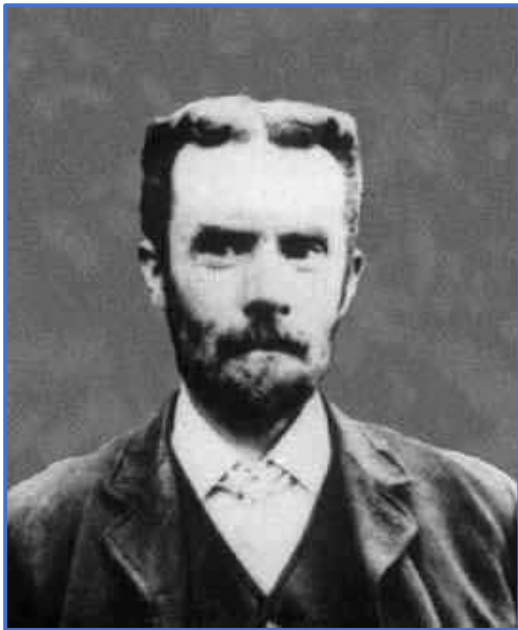
Pierre-Simon, Marquis de Laplace



- 1749–1827 French mathematician and astronomer
- **Laplace**, along with **Lagrange** and **Legendre** of his time, was known as one of the "Three L's" of France
- In 1779, he introduced his transformation method to **solve differential equations**, but it was not until a century later that this method gained widespread recognition and popularity

1. Introduction of Laplace Transform

Oliver Heaviside



- 1850–1925 British engineer
- Self-taught with only elementary education
- First to derive the **distortionless transmission line solution**
- First to write **Maxwell's equations in their modern form**
- In the late 19th century, developed the "operational calculus," which coincidentally aligned with the Laplace transform

1. Introduction of Laplace Transform

Advantages:

- **Simplifies problem-solving**, especially for differential equations of systems, **as initial conditions are automatically incorporated**, making it more widely applicable
- Converts differentiation/integration into multiplication/division: Exponential and transcendental functions are transformed into elementary algebraic expressions
- Establishes the concept of **transfer function** based on the convolution theorem, facilitating system analysis

Disadvantages:

- The **physical** interpretation is **less intuitive** compared to the Fourier transform

1. Introduction of Laplace Transform

Fundamentals

- **Direct Transform**
- **Inverse Transform**
- **Properties**

Applications

- **Solving Differential Equations**
- **Analyzing Dynamic Circuits**
- **System Function: Analyzing system characteristics and block diagrams**

2. Definition of Laplace Transform

Main Content

1. Definition of Laplace Transform
2. Region of Convergence (ROC)

2. Definition of Laplace Transform

1. Definition of Laplace Transform

Fourier Transform

$$\mathcal{F}[f(t)] = \int_{-\infty}^{\infty} f(t) \cdot e^{-j\omega t} dt$$

$f(t)$ may not satisfy the absolute integrability condition

$f(t)$ is multiplied by an exponential decay factor $e^{-\sigma t}$, σ is a real number

By choosing an appropriate value for σ , $f(t)e^{-\sigma t}$ can more easily meet the absolute integrability requirement

2. Definition of Laplace Transform

1. Definition of Laplace Transform

$$\begin{aligned}\mathcal{F}\left[f(t) \cdot e^{-\sigma t}\right] &= \int_{-\infty}^{\infty} \left[f(t) e^{-\sigma t}\right] \cdot e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} f(t) \cdot e^{-(\sigma + j\omega)t} dt \\ &= F(\sigma + j\omega)\end{aligned}$$

Let $\sigma + j\omega = s$, has the dimension of frequency and is referred to as the complex frequency, then

$$\begin{aligned}F(s) &= \mathcal{L}[f(t)] \\ &= \int_{-\infty}^{\infty} f(t) e^{-st} dt\end{aligned}$$

2. Definition of Laplace Transform

1. Definition of Laplace Transform

Laplace Transform (Forward Transform)

$$F(s) = \mathcal{L}[f(t)] = \mathcal{F}[f(t) \cdot \mathbf{e}^{-\sigma t}] = F(\sigma + \mathbf{j}\omega)$$

Inverse Fourier transform $\mathbf{F}(\sigma + \mathbf{j}\omega)$ yields $\mathbf{f}(t) \cdot \mathbf{e}^{-\sigma t}$

$$\begin{aligned} f(t) \mathbf{e}^{-\sigma t} &= \mathcal{F}^{-1}[F(\sigma + \mathbf{j}\omega)] \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\sigma + \mathbf{j}\omega) \mathbf{e}^{\mathbf{j}\omega t} \mathbf{d}\omega \end{aligned}$$

Multiplying both sides by $\mathbf{e}^{\sigma t}$ gives the **inverse Laplace transform**:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\sigma + \mathbf{j}\omega) \mathbf{e}^{(\sigma + \mathbf{j}\omega)t} \mathbf{d}\omega$$

2. Definition of Laplace Transform

1. Definition of Laplace Transform

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\sigma + j\omega) e^{(\sigma + j\omega)t} d\omega$$

By substituting $s = \sigma + j\omega$, we obtain:

$$f(t) = \frac{1}{2\pi j} \int_{\sigma - j\infty}^{\sigma + j\infty} F(s) e^{st} ds$$

σ is a constant

Integration limits: for ω is $\int_{-\infty}^{\infty}$, for s is $\int_{\sigma - j\infty}^{\sigma + j\infty}$

The **path of integration** is a vertical line within the region of convergence (ROC)

2. Definition of Laplace Transform

1. Definition of Laplace Transform

$$\begin{cases} F(s) = \mathcal{L}[f(t)] = \int_{-\infty}^{\infty} f(t) e^{-st} dt & \sigma? \\ f(t) = \mathcal{L}^{-1}[F(s)] = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s) e^{st} ds \end{cases}$$

**Bilateral (Two-sided)
Laplace Transform**

**Inverse Laplace
Transform**

These equations represent the generalized Fourier transform, also known as the **bilateral Laplace transform**

Where $s = \sigma + j\omega$, σ a predetermined real constant, ω is the frequency variable

Notation: $f(t) \xleftrightarrow{LT} F(s)$

2. Definition of Laplace Transform

2. Region of Convergence (ROC)

$s = \sigma + j\omega$, σ **predetermined real constant**

σ must be chosen such that the product $f(t) \cdot e^{-\sigma t}$ converges absolutely

Region of Convergence: the set of **all s values** for which the Laplace transform $F(s)$ exists

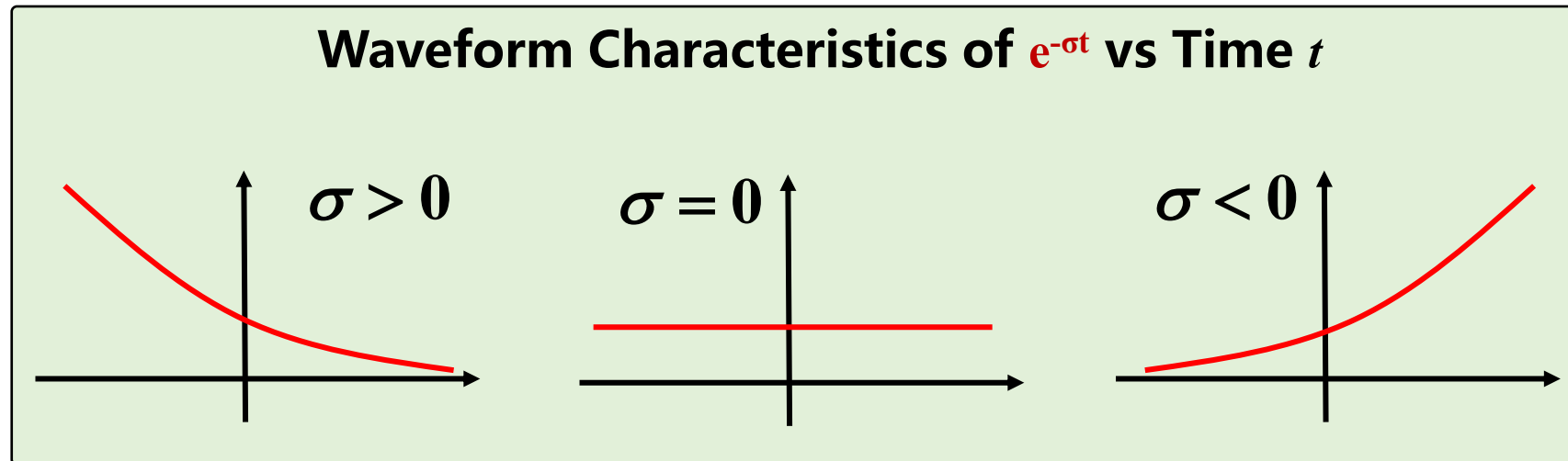
Notation: Abbreviated as **ROC** (region of convergence)

Defines the existence condition for the Laplace transform

2. Definition of Laplace Transform

2. Region of Convergence (ROC)

When analyzing a signal $f(t)$, the Laplace transform applies an exponential damping factor $e^{-\sigma t}$. The behavior of this damping factor depends on the value of σ , and attenuation is unidirectional



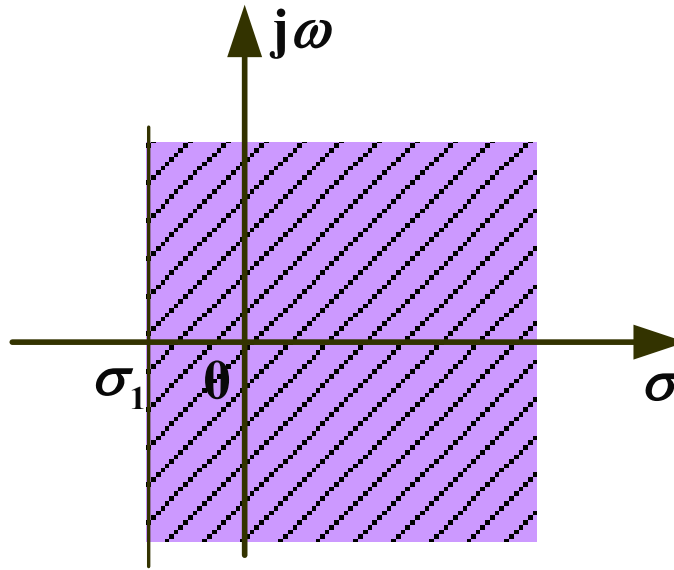
2. Definition of Laplace Transform

2. Region of Convergence (ROC)

(1) If $f(t)$ is a **right-sided** signal, and for all real numbers $\sigma > \sigma_1$

$$\lim_{t \rightarrow \infty} f(t) \cdot e^{-\sigma t} = 0$$

$$\text{ROC: } \text{Re}[s] = \sigma > \sigma_1$$



Complex plane (**s plane**): $s = \sigma + j\omega$

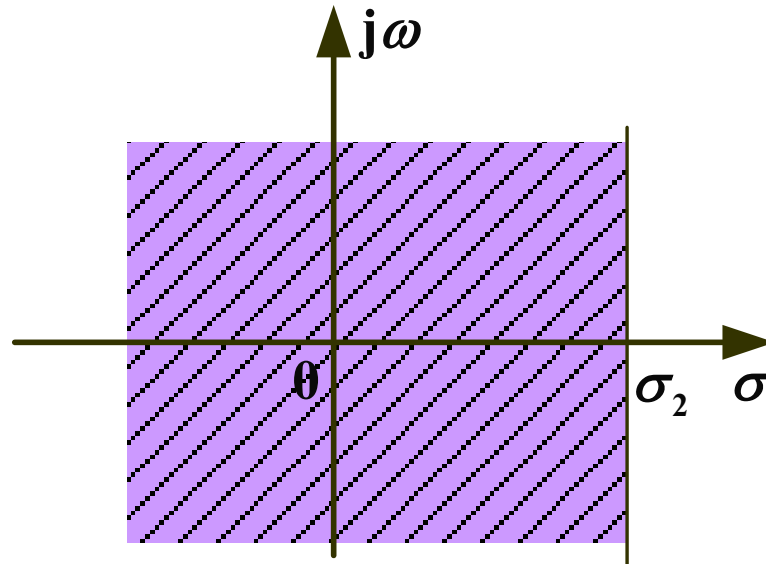
2. Definition of Laplace Transform

2. Region of Convergence (ROC)

(2) If $f(t)$ is a **left-sided** signal, and for all real numbers $\sigma < \sigma_2$

$$\lim_{t \rightarrow -\infty} f(t) \cdot e^{-\sigma t} = 0$$

$$\text{ROC: } \text{Re}[s] = \sigma < \sigma_2$$

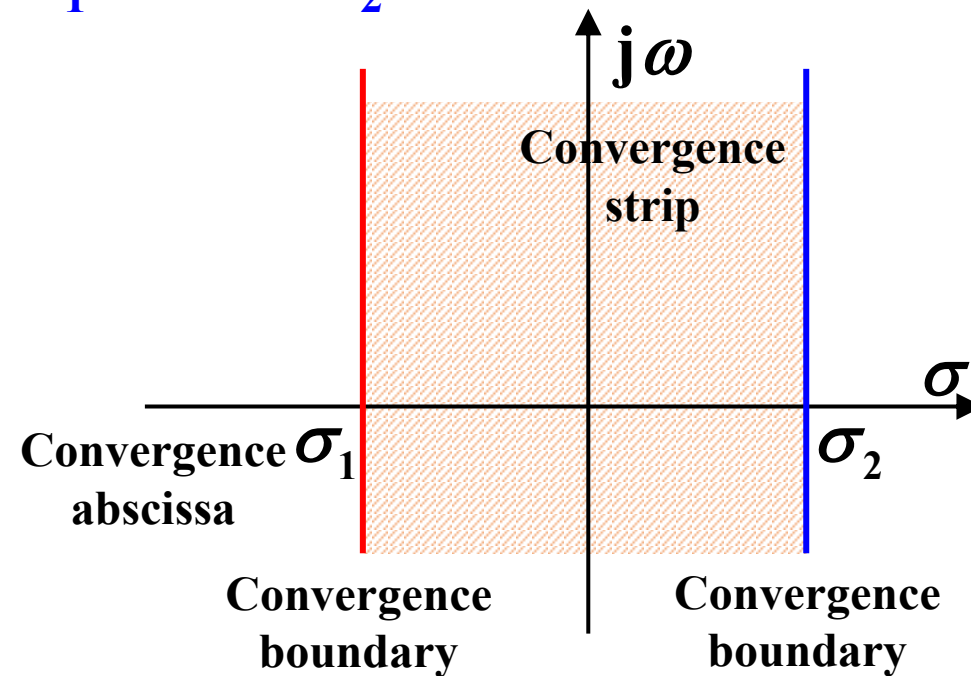


2. Definition of Laplace Transform

2. Region of Convergence (ROC)

(3) For a two-sided signal, the ROC is the intersection of the ROCs from two one-sided signals

$$\text{ROC: } \sigma_1 < \sigma < \sigma_2$$



2. Definition of Laplace Transform

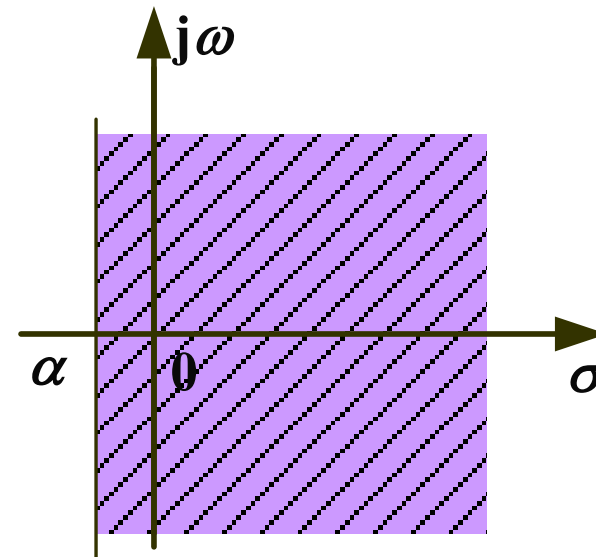
Example 1

Determine the ROC for the signal $f(t) = e^{\alpha t} u(t)$, where $\alpha < 0$

$$\lim_{t \rightarrow \infty} e^{\alpha t} \cdot e^{-\sigma t} = \lim_{t \rightarrow \infty} e^{(\alpha - \sigma)t} = 0$$

$$\alpha - \sigma < 0$$

$$\text{ROC: } \sigma > \alpha$$



2. Definition of Laplace Transform

Example 2

Find the ROC for the Laplace transform of the signal, $f(t) = e^{t^2} u(t)$

$$\lim_{t \rightarrow \infty} e^{t^2} \cdot e^{-\sigma t} = \lim_{t \rightarrow \infty} e^{(t^2 - \sigma t)} \neq 0$$

The Laplace transform does not exist (**diverges for all σ**)

2. Definition of Laplace Transform

2. Region of Convergence (ROC)

A signal is said to be of exponential order if it satisfies

$$\begin{cases} \lim_{t \rightarrow \infty} f(t) e^{-\sigma t} = 0 \quad (\sigma > \sigma_1) \\ \lim_{t \rightarrow -\infty} f(t) e^{-\sigma t} = 0 \quad (\sigma < \sigma_2) \end{cases}$$

The Laplace transform always exists for **bounded aperiodic** signals

For power functions t^n , which grow slower than exponentials e^t , $\lim_{t \rightarrow \infty} t^n e^{-\sigma t} = 0, \quad \sigma > 0$

For unilateral exponential signals $e^{\alpha t}$, $\lim_{t \rightarrow \infty} e^{\alpha t} e^{-\sigma t} = 0, \quad \sigma > \alpha$

Signals like e^{t^2} , which grow faster than exponentials, have no ROC (non-exponential order) and thus no Laplace transform

2. Definition of Laplace Transform

Summary

Considering that actual signals are **causal signals** and physically realizable systems are **causal**, this course primarily focuses on the **right-sided unilateral Laplace transform**

The region of convergence (ROC) **does not include any poles**, and generally, there **is no need to explicitly note the ROC**

$$\begin{cases} F(s) = \mathcal{L}[f(t)] = \int_{0-}^{\infty} f(t) e^{-st} dt \\ f(t) = \mathcal{L}^{-1}[F(s)] = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s) e^{st} ds \end{cases}$$

Here, $f(t)$ is referred to as the **original function**, and $F(s)$ is referred to as the **image function**

Main Content

Laplace Transform of Common Signals

1. Step Function
2. Exponential Function
3. Unit Impulse Signal
4. $t^n u(t)$

3. Laplace Transform of Common Signals

1. Step Function

$$\begin{aligned}\mathcal{L}[u(t)] &= \int_{0_-}^{\infty} 1 \cdot e^{-st} dt \\ &= \frac{1}{-s} e^{-st} \Big|_{0_-}^{\infty} = \frac{1}{s}\end{aligned}$$

$$\text{Let } \lim_{t \rightarrow \infty} e^{-\sigma t} = 0$$

$$\text{ROC: } \operatorname{Re}[s] > 0 \text{ or } \sigma > 0$$

2. Exponential Function

$$\mathcal{L}\left[e^{-s_0 t} u(t)\right] = \int_{0_-}^{\infty} e^{-s_0 t} \mathbf{e}^{-\mathbf{s}t} dt \quad (s_0 = \alpha + j\omega_0)$$

$$= \frac{e^{-(s_0 + s)t}}{-(s_0 + s)} \Big|_{0_-}^{\infty} = \frac{\mathbf{1}}{\mathbf{s}_0 + \mathbf{s}}$$

$$\text{Let } \lim_{\mathbf{t} \rightarrow \infty} \mathbf{e}^{-\mathbf{s}_0 t} \cdot \mathbf{e}^{-\mathbf{s}t} = 0 \quad \text{thus } \lim_{\mathbf{t} \rightarrow \infty} e^{-(\alpha + \sigma)t} = 0$$

$$\text{ROC: } \sigma > -\alpha$$

3. Laplace Transform of Common Signals

3. Unit Impulse Signal

$$\begin{aligned}\mathcal{L}[\delta(t)] &= \int_{0_-}^{\infty} \delta(t) \cdot e^{-st} dt \\ &= 1\end{aligned}$$

ROC: The entire s -plane converges

3. Laplace Transform of Common Signals

4. $t^n u(t)$

Integration
by parts

$$\begin{aligned}\mathcal{L}[t^n u(t)] &= \int_{0_-}^{\infty} t^n \cdot e^{-st} dt \\ &= \int_{0_-}^{\infty} t^n \frac{1}{-s} de^{-st} \\ &= \frac{e^{-st}}{-s} t^n \bigg|_{0_-}^{\infty} - \frac{n}{-s} \int_{0_-}^{\infty} t^{n-1} e^{-st} dt = \frac{n}{s} \int_{0_-}^{\infty} t^{n-1} e^{-st} dt\end{aligned}$$

$$\text{Let } \lim_{t \rightarrow \infty} t^n e^{-\sigma t} = 0 \quad \text{ROC: } \sigma > 0$$

$$\text{Thus } \mathcal{L}[t^n u(t)] = \frac{n}{s} \mathcal{L}[t^{n-1} u(t)]$$

3. Laplace Transform of Common Signals

4. $t^n u(t)$

$$\mathcal{L}[t^n u(t)] = \frac{n}{s} \mathcal{L}[t^{n-1} u(t)]$$

$$n=0 \quad t^n u(t) = u(t)$$

$$\mathcal{L}[u(t)] = \frac{1}{s}$$

$$n=1$$

$$\begin{aligned} \mathcal{L}[tu(t)] &= \frac{1}{s} \left(\frac{1}{s} \right) \\ &= \frac{1}{s^2} \end{aligned}$$

$$\text{Thus } \mathcal{L}[t^n u(t)] = \frac{n!}{s^{n+1}}$$

$$n=2$$

$$\mathcal{L}[t^2 u(t)] = \frac{2}{s} \mathcal{L}[tu(t)] = \frac{2}{s} \cdot \frac{1}{s^2} = \frac{2}{s^3}$$

$$n=3$$

$$\mathcal{L}[t^3 u(t)] = \frac{3}{s} \mathcal{L}[t^2 u(t)] = \frac{3}{s} \cdot \frac{2}{s^3} = \frac{6}{s^4}$$

.....

$$\text{ROC: } \sigma > 0$$

Summary

Laplace Transform of Common Signals

1. Step Function
2. Exponential Function
3. Unit Impulse Signal
4. $t^n u(t)$

Main Content

1. Linearity Property
2. Shifting Property
3. s-Domain Shift Property
4. Scaling Property
5. Convolution Theorem
6. Differentiation Property
7. Integration Property
8. Initial Value Theorem
9. Final Value Theorem
10. s-Domain Differentiation Property
11. s-Domain Integration Property

Note: Unless otherwise specified, all refer to **unilateral** transforms

1. Linearity Property

IF $\mathcal{L}[f_1(t)] = F_1(s)$

$$\mathcal{L}[f_2(t)] = F_2(s)$$

K_1, K_2 constants, thus

$$\mathcal{L}[K_1 f_1(t) + K_2 f_2(t)] = K_1 F_1(s) + K_2 F_2(s)$$

- The ROC is the **intersection** of the individual ROCs
- The ROC does not include any poles

4. Basic Properties of Laplace Transform

Example 1

$$f(t) = \cos(\omega_0 t) u(t) = \frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t}) u(t)$$

Given

$$\mathcal{L}[e^{-s_0 t} u(t)] = \frac{1}{s + s_0} \quad (s_0 \text{ is a complex number})$$

Thus

$$\mathcal{L}[\cos(\omega_0 t) u(t)] = \frac{1}{2} \left(\frac{1}{s - j\omega_0} + \frac{1}{s + j\omega_0} \right) = \frac{s}{s^2 + \omega_0^2}$$

Similarly

$$\mathcal{L}[\sin(\omega_0 t) u(t)] = \frac{\omega_0}{s^2 + \omega_0^2}, (\sigma > 0)$$

4. Basic Properties of Laplace Transform

Example 1

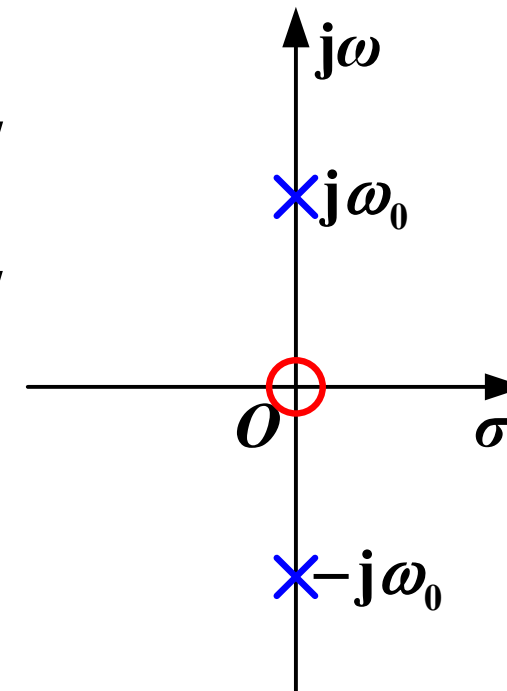
$$\cos(\omega_0 t)u(t) \leftrightarrow F(s) = \frac{s}{s^2 + \omega_0^2}$$

Pole-Zero Plot

- The roots of the **denominator** polynomial of $F(s)$, denoted as $p_1, p_2, \dots, p_n$, are called the **poles** of $F(s)$
- The roots of the **numerator** polynomial of $F(s)$, denoted as $z_1, z_2, \dots, z_n$, are called the **zeros** of $F(s)$

On the s -plane, plot the pole-zero diagram of $F(s)$:

- **Poles:** Represented by $\times$ (cross marks)
- **Zeros:** Represented by $\circ$ (open circles)



2. Shifting Property

If $L[f(t)] = F(s)$,

Then the Laplace transform of $f(t - t_0)u(t - t_0)$ (where $t_0 > 0$) is:

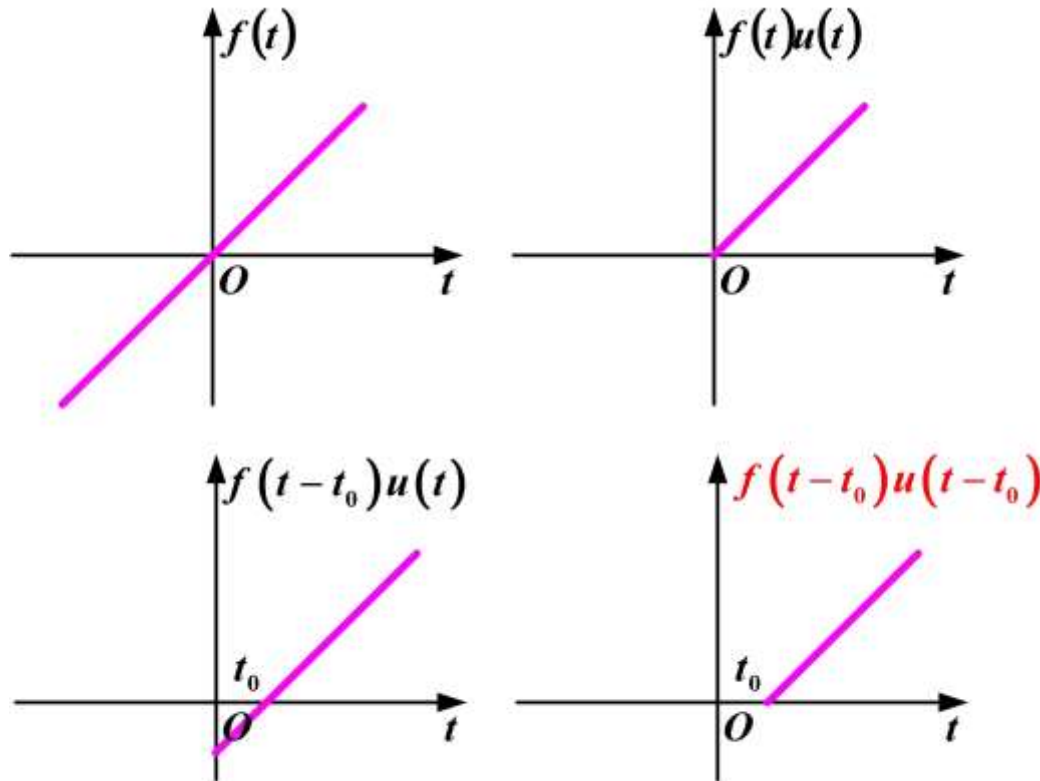
$$F(s)e^{-st_0}$$

$$f(t) \xrightarrow{\text{Unilateral Transform}} f(t)u(t) \xrightarrow{\text{Time Shift}} f(t - t_0)u(t - t_0)$$

$$\left\{ \begin{array}{l} t \rightarrow t - t_0 \\ \text{determine time shift based on } u(t) \end{array} \right.$$

4. Basic Properties of Laplace Transform

2. Shifting Property



$\left\{ \begin{array}{l} t \rightarrow t - t_0 \\ \text{determine time shift based on } u(t) \end{array} \right.$

4. Basic Properties of Laplace Transform

2. Shifting Property

Proof $\mathcal{L}[f(t-t_0)u(t-t_0)] = \int_{0_-}^{\infty} f(t-t_0)u(t-t_0)e^{-st} dt$

Let $\tau = t - t_0$ $= \int_{t_0}^{\infty} f(t-t_0)e^{-st} dt$

Then $t = \tau + t_0$ $= e^{-st_0} \int_{0_-}^{\infty} f(\tau)e^{-s\tau} d\tau$

$dt = d\tau$, **Substitute into the above equation** $= F(s)e^{-st_0}$

4. Basic Properties of Laplace Transform

Example 2

Find the Laplace transform of $f(t) = e^{-t}u(t-2)$

$$f(t) = e^{-2}e^{-(t-2)}u(t-2)$$

Given $e^{-t}u(t) \leftrightarrow \frac{1}{s+1}$

$$F(s) = e^{-2} \cdot \frac{1}{s+1} \cdot e^{-2s}$$

$$= \frac{1}{s+1} \cdot e^{-2(s+1)}$$

3. s-Domain Shift Property

If $\mathcal{L}[f(t)] = F(s)$,

Then $\mathcal{L}[f(t)e^{-\alpha t}] = F(s + \alpha)$

Proof

$$\begin{aligned}\mathcal{L}[f(t)e^{-\alpha t}] &= \int_{0_-}^{\infty} f(t)e^{-\alpha t}e^{-st} dt \\ &= F(s + \alpha)\end{aligned}$$

4. Basic Properties of Laplace Transform

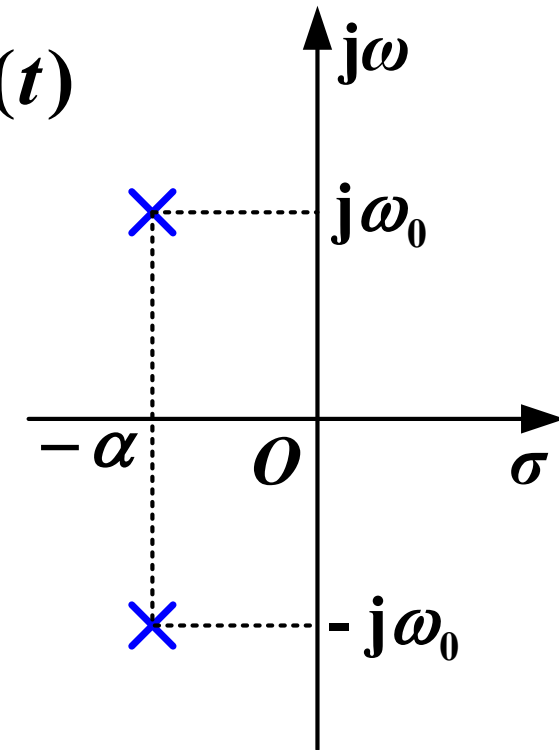
Example 3

Find the Laplace transform of $e^{-\alpha t} \cos(\omega_0 t) u(t)$

Given: $\mathcal{L}[\cos(\omega_0 t) u(t)] = \frac{s}{s^2 + \omega_0^2}$

Thus $e^{-\alpha t} \cos(\omega_0 t) u(t) \xleftrightarrow{LT} \frac{s + \alpha}{(s + \alpha)^2 + \omega_0^2}$

Similarly : $e^{-\alpha t} \sin(\omega_0 t) u(t) \xleftrightarrow{LT} \frac{\omega_0}{(s + \alpha)^2 + \omega_0^2}$



ω_0 : angular frequency
 $-\alpha$: amplitude

4. Scaling Property

$$\text{If } \mathcal{L}[f(t)] = F(s), \quad \text{Then } \mathcal{L}[f(\textcolor{red}{a}t)] = \frac{1}{\textcolor{red}{a}} F\left(\frac{s}{\textcolor{red}{a}}\right) \quad (\textcolor{red}{a} > 0)$$

When both time shifting and time scaling are involved:

$$\mathcal{L}[f(at - b)u(at - b)] = \frac{1}{a} F\left(\frac{s}{a}\right) e^{-s\frac{b}{a}} \quad (a > 0, b > 0)$$

4. Basic Properties of Laplace Transform

5. Convolution Theorem

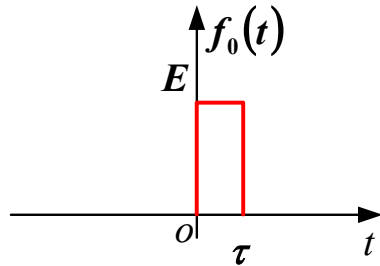
If $\mathcal{L}[f_1(t)] = F_1(s)$, $\mathcal{L}[f_2(t)] = F_2(s)$, $f_1(t), f_2(t)$ are causal signals, then:

$$\mathcal{L}[f_1(t) * f_2(t)] = F_1(s)F_2(s)$$

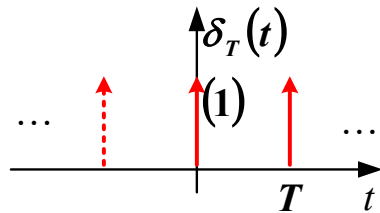
$$\mathcal{L}[f_1(t) \cdot f_2(t)] = \frac{1}{2\pi j} F_1(s) * F_2(s)$$

4. Basic Properties of Laplace Transform

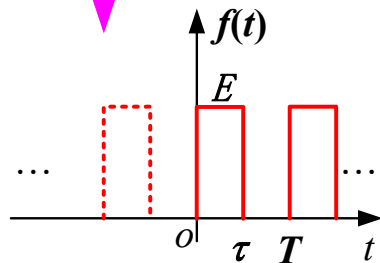
Example 4



$$\begin{aligned}\mathcal{L}[\delta_T(t)] &= 1 + e^{-sT} + e^{-s2T} + \dots \\ &= \frac{1}{1 - e^{-sT}}\end{aligned}$$



Convolution



For a general signal $f(t) = f_0(t) * \delta_T(t)$

$$F(s) = F_0(s) \frac{1}{1 - e^{-sT}}$$

$\frac{1}{1 - e^{-sT}}$	Termed the "Periodization Factor"
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6. Differentiation Property

(1) Differentiation Property for Bilateral Laplace Transform

$$\text{If } f(t) \xleftrightarrow{\mathcal{L}} F(s), \text{ then } \frac{d f(t)}{d t} \xleftrightarrow{\mathcal{L}} s F(s)$$

Proof:

The **inverse** bilateral Laplace transform is given by:

$$f(t) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s) e^{st} ds$$

Differentiating both sides with respect to t :

$$\frac{d f(t)}{d t} = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s) s e^{st} ds$$

Application: System function analysis

6. Differentiation Property

(2) Differentiation Property for Unilateral Laplace Transform (for reference)

$$\text{If } f(t) \xleftrightarrow{\mathcal{L}} F(s), \text{ then } \frac{d f(t)}{d t} \xleftrightarrow{\mathcal{L}} sF(s) - f(0_-)$$

Proof:

$$\begin{aligned} \int_{0_-}^{\infty} f'(t) e^{-st} dt &= \int_{0_-}^{\infty} e^{-st} df(t) \\ &= f(t) e^{-st} \Big|_{0_-}^{\infty} - \int_{0_-}^{\infty} f(t) d e^{-st} \\ &= -f(0_-) - \int_{0_-}^{\infty} (-s) f(t) e^{-st} dt \\ &= -f(0_-) + sF(s) \end{aligned}$$

6. Differentiation Property

$$\frac{d f(t)}{d t} \xleftrightarrow{\mathcal{L}} s F(s) - f(0_-)$$

$$\begin{aligned} \frac{d f^2(t)}{d t} &\xleftrightarrow{\mathcal{L}} s \left[s F(s) - f(0_-) \right] - f'(0_-) \\ &= s^2 F(s) - s f(0_-) - f'(0_-) \end{aligned}$$

$$\frac{d f^n(t)}{d t} \xleftrightarrow{\mathcal{L}} s^n F(s) - \sum_{r=0}^{n-1} s^{n-r-1} f^{(r)}(0_-)$$

Application:

- (1) Solving differential equation
- (2) Dynamic circuit analysis

7. Integration Property

(1) Integration Property for Bilateral Laplace Transform

If $f(t) \xleftrightarrow{\mathcal{L}} F(s)$, then $\int_{-\infty}^t f(\tau) d\tau \xleftrightarrow{\mathcal{L}} \frac{F(s)}{s}$

Proof:

$$\int_{-\infty}^t f(\tau) d\tau = f(t) * u(t)$$

$$u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s}$$

Using the time-domain convolution theorem, obtain

$$\int_{-\infty}^t f(\tau) d\tau \xleftrightarrow{\mathcal{L}} \frac{F(s)}{s}$$

7. Integration Property

(2) Integration Property for Unilateral Laplace Transform (for reference)

If $f(t) \xleftrightarrow{\mathcal{L}} F(s)$, then

$$\int_{-\infty}^t f(\tau) d\tau \xleftrightarrow{\mathcal{L}} \frac{F(s)}{s} + \frac{f^{(-1)}(0_-)}{s}$$

Proof:

$$\int_{-\infty}^t f(\tau) d\tau = \underbrace{\int_{-\infty}^{0_-} f(\tau) d\tau}_{\textcircled{1}} + \underbrace{\int_{0_-}^t f(\tau) d\tau}_{\textcircled{2}}$$

$$\textcircled{1} \quad f^{(-1)}(t) = \int_{-\infty}^t f(\tau) d\tau \quad \text{ie} \quad \int_{-\infty}^{0_-} f(\tau) d\tau = f^{(-1)}(0_-)$$

So, Laplace transform of $\textcircled{1}$ is $\mathcal{L}\left[f^{(-1)}(0_-)\right] = \frac{f^{(-1)}(0_-)}{s}$

7. Integration Property

$$\int_{-\infty}^t f(\tau) d\tau = \underbrace{\int_{-\infty}^{0-} f(\tau) d\tau}_{\textcircled{1}} + \underbrace{\int_{0-}^t f(\tau) d\tau}_{\textcircled{2}}$$

The LT of $\textcircled{2}$ is $\int_{0-}^{\infty} \left[\int_{0-}^t f(\tau) d\tau \right] e^{-st} dt$

$$= \int_{0-}^{\infty} \left[\int_{0-}^{\infty} f(\tau) u(t-\tau) d\tau \right] e^{-st} dt$$

$$= \int_{0-}^{\infty} f(\tau) \int_{0-}^{\infty} u(t-\tau) e^{-st} dt d\tau$$

$$= \frac{1}{s} \int_{0-}^{\infty} f(\tau) e^{-s\tau} d\tau = \frac{F(s)}{s}$$

8. Initial Value Theorem

If $f(t)$ and $\frac{df(t)}{dt}$ are Laplace-transformable, and $f(t) \leftrightarrow F(s)$, then:

$$\lim_{t \rightarrow 0_+} f(t) = f(0_+) = \lim_{s \rightarrow \infty} sF(s)$$

Notes:

1. If $F(s)$ is not a proper fraction, express it as a polynomial plus a proper fraction:

$$F(s) = F_1(s) + \cdots + k_1 s + k_0$$

2. The initial value is determined by:

$$f(0_+) = \lim_{s \rightarrow \infty} sF_1(s)$$

3. For the polynomial part:

$$f_2(t) = \cdots k_1 s + k_0 \leftrightarrow \cdots + k_1 \delta'(t) + k_0 \delta(t)$$

$$f_2(0_+) = 0$$

$$\text{But } \lim_{s \rightarrow \infty} s(\cdots + k_1 s + k_0) = \infty$$

8. Initial Value Theorem

Proof:

From the time-domain differentiation property:

Using the 0_- system:
$$\int_{0_-}^{\infty} \frac{d f(t)}{d t} e^{-s t} d t = s F(s) - f(0_-)$$

Using the 0_+ system:
$$\int_{0_+}^{\infty} \frac{d f(t)}{d t} e^{-s t} d t = s F(s) - f(0_+)$$

$$\lim_{s \rightarrow \infty} \left[\int_{0_+}^{\infty} \frac{d f(t)}{d t} e^{-s t} d t \right] = \int_{0_+}^{\infty} \frac{d f(t)}{d t} \left[\lim_{s \rightarrow \infty} e^{-s t} \right] d t = 0$$

$$\therefore f(0_+) = \lim_{s \rightarrow \infty} s F(s)$$

4. Basic Properties of Laplace Transform

Example 5

$$F(s) = \frac{2s}{s+1}, \quad f(0_+) = ?$$

Because $F(s) = \frac{2s}{s+1} = -\frac{2}{s+1} + 2$

The function $f(t)$
contains a term $2\delta(t)$

Let $F_0(s) = -\frac{2}{s+1}$

Thus $f(0_+) = \lim_{s \rightarrow \infty} [sF_0(s)]$

$$= \lim_{s \rightarrow \infty} s \frac{-2}{s+1} = \lim_{s \rightarrow \infty} \frac{-2}{1 + \frac{1}{s}} = -2$$

9. Final Value Theorem

Let $f(t)$ and $\frac{d}{dt}f(t)$ have Laplace transforms, and suppose:

$$L[f(t)] = F(s)$$

Then:

$$f(\infty) = \lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$

Conditions for Finite Final Value:

1. All poles in the **left** half-plane (LHP) of s : Final value is 0 (system stabilizes to zero)
2. Single pole at $s = 0$: The final value equals the coefficient k of the $\frac{k}{s}$ term in the **partial fraction expansion**

9. Final Value Theorem

Proof:

Based on the formula derived from the proof of the Initial Value Theorem

$$sF(s) = f(0_+) + \int_{0_+}^{\infty} \frac{d f(t)}{d t} e^{-st} d t$$

$$\lim_{s \rightarrow 0} sF(s) = f(0_+) + \lim_{s \rightarrow 0} \int_{0_+}^{\infty} \frac{d f(t)}{d t} e^{-st} d t$$

$$= f(0_+) + \lim_{t \rightarrow \infty} f(t) - f(0_+)$$

$$= \lim_{t \rightarrow \infty} f(t)$$

9. Final Value Theorem

Conditions for the Existence of a Final Value

When poles are in the **left** half-plane (LHP) of s , the final value is **0**

Single Pole $\frac{1}{s+2} \leftrightarrow e^{-2t} u(t)$

Complex Conjugate Poles $\frac{s+2}{(s+2)^2 + \omega_1^2} \leftrightarrow e^{-2t} \cos(\omega_1 t) u(t)$

Double Pole $\frac{1}{(s+2)^2} \leftrightarrow te^{-2t} u(t)$

When poles are in the right half-plane (RHP) of s , the original function diverges $\frac{1}{s-2} \leftrightarrow e^{+2t} u(t)$

9. Final Value Theorem

Conditions for the Existence of a Final Value

When poles are on the **axis** of the s -plane

Single pole at origin $\frac{1}{s} \leftrightarrow u(t)$ Final value is 1

Multiple poles at origin $\frac{1}{s^2} \leftrightarrow tu(t)$ No finite final value

Complex conjugate poles on imaginary axis $\frac{s}{s^2 + \omega_1^2} \leftrightarrow \cos(\omega_1 t)u(t)$ Oscillates indefinitely (no final value)

9. Final Value Theorem

There are only two cases where a finite final value exists

- (1) All poles in the left half-plane (LHP) of s , Final value = 0
- (2) Single pole at $s = 0$, The final value equals the coefficient k of the k/s term in the partial fraction expansion, calculated as:

$ku(t)$

$$k = sF(s) \Big|_{s=0}$$

$$f(\infty) = \lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$

10. s-Domain Differentiation Property

$$\text{If } \mathcal{L}[f(t)] = F(s), \text{ then } \mathcal{L}[tf(t)] = -\frac{dF(s)}{ds}$$

Generalization: $\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n F(s)}{ds^n}$

Summary

1. **Linearity Property**
2. **Shifting Property**
3. **s-Domain Shift Property**
4. **Scaling Property**
5. **Convolution Theorem**
6. **Differentiation Property**
7. **Integration Property**
8. **Initial Value Theorem**
9. **Final Value Theorem**
10. **s-Domain Differentiation Property**

4. Basic Properties of Laplace Transform

Summary

Signal	Time domain	S-domain
Step function	$u(t)$	$\frac{1}{s}$
exponential function	$e^{-s_0 t} u(t)$	$\frac{1}{s + s_0}$
Unit impulse signal	$\delta(t)$	1
$tu(t)$	$t^n u(t)$	$\frac{n!}{s^{n+1}}$
cos	$\cos(w_0 t) u(t)$	$\frac{s}{s^2 + w_0^2}$
sin	$\sin(w_0 t) u(t)$	$\frac{w_0}{s^2 + w_0^2}$

4. Basic Properties of Laplace Transform

Summary

Property	$f(t)$	$F(s)$
Linearity	$K_1 f_1(t) + K_2 f_2(t)$	$K_1 F_1(s) + K_2 F_2(s)$
Time shifting	$f(t - t_0)u(t - t_0)$	$F(s)e^{-st_0}$
s-domain shift	$f(t)e^{-\alpha t}$	$F(s + \alpha)$
Scaling	$f(at)$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
Time shifting + scaling	$f(at - b)u(at - b)$	$\frac{1}{a}F\left(\frac{s}{a}\right)e^{-s\frac{b}{a}}$
Convolution	$f_1(t) * f_2(t)$	$F_1(s) \cdot F_2(s)$
Bilateral differentiation	$\frac{df(t)}{dt}$	$sF(s)$

4. Basic Properties of Laplace Transform

Summary

Unilateral differentiation	$\frac{df(t)}{dt}$	$sF(s) - f(0_-)$
Bilateral integral	$\int_{-\infty}^t f(\tau) d\tau$	$\frac{F(s)}{s}$
Unilateral integral	$\int_{-\infty}^t f(\tau) d\tau$	$\frac{F(s)}{s} + \frac{f^{-1}(0_-)}{s}$
Initial value	$f(0_+)$	$\lim_{s \rightarrow \infty} sF(s)$
Final value	$f(\infty)$	$\lim_{s \rightarrow 0} sF(s)$
s-domain differentiation	$t^n f(t)$	$(-1)^n \frac{d^n F(s)}{ds^n}$
s-domain integration	$\frac{f(t)}{t}$	$\int_s^\infty F(\lambda) d\lambda$

Main Content

1. Inverse Laplace Transform
2. Single Real Poles
3. Complex Conjugate Poles
4. Multiple-Order Poles

Key Focus: Partial fraction expansion method for inverse Laplace transforms

Critical Challenges: Handling complex conjugate poles and repeated roots

1. Inverse Laplace Transform

1. **Complex Contour Integration via Definition** - Uses the residue theorem to evaluate contour integrals
2. **Partial Fraction Expansion — Table Lookup Method (Key Technique)**
 - **Decomposes $F(s)$ into simpler fractions, then matches them to time-domain functions via standard tables**
3. **Computer-Aided Solutions**
 - **Implemented in tools like MATLAB**

5. Inverse Laplace Transform

(1) Partial Fraction Expansion

$$\mathcal{L}[\delta(t)] = 1$$

$$\mathcal{L}[\delta^{(n)}(t)] = s^n$$

Standard Signal
Transform Table

$$\mathcal{L}[u(t)] = \frac{1}{s} \quad \mathcal{L}[t] = \frac{1}{s^2} \quad \mathcal{L}[t^n] = \frac{n!}{s^{n+1}}$$

$$\mathcal{L}[\cos(\omega_0 t)u(t)] = \frac{s}{s^2 + \omega_0^2} \quad \mathcal{L}[\sin(\omega_0 t)u(t)] = \frac{\omega_0}{s^2 + \omega_0^2}$$

$$\mathcal{L}[e^{-\alpha t}u(t)] = \frac{1}{s + \alpha} \quad \mathcal{L}[e^{-\alpha t}\cos(\omega_0 t)u(t)] = \frac{s + \alpha}{(s + \alpha)^2 + \omega_0^2}$$

$$\mathcal{L}[e^{-\alpha t}\sin(\omega_0 t)u(t)] = \frac{\omega_0}{(s + \alpha)^2 + \omega_0^2}$$

(1) Partial Fraction Expansion

Typically, $F(s)$ is expressed as a rational function:

$$F(s) = \frac{B(s)}{A(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0}$$

a_i, b_i are real coefficients, m, n are positive integers

If $m < n$, $F(s)$ is a proper rational function

Factorization

$$F(s) = \frac{B(s)}{A(s)} = \frac{b_m (s - z_1)(s - z_2) \cdots (s - z_m)}{a_n (s - p_1)(s - p_2) \cdots (s - p_n)}$$

Zeros: $z_1, z_2, z_3 \cdots z_m$ are roots of $B(s) = 0$

Poles: $p_1, p_2, p_3 \cdots p_n$ are roots of $A(s) = 0$

5. Inverse Laplace Transform

(1) Partial Fraction Expansion

Step-by-Step Procedure:

- Identify Poles of $F(s)$
- Expand $F(s)$ into Partial Fractions
- Inverse Transform $f(t)$ via Laplace Table

5. Inverse Laplace Transform

(1) Partial Fraction Expansion

Proper Rational Function: $F(s) = \frac{B(s)}{(s - p_1)(s - p_2) \cdots (s - p_n)}$

Case 1: Distinct Real Poles

$p_1, p_2, p_3 \dots p_n$ are distinct real roots

$$\frac{1}{(s - p_1)(s - p_2)} = \frac{k_1}{s - p_1} + \frac{k_2}{s - p_2}$$

$$\frac{k}{s - p} \leftrightarrow e^{pt} u(t)$$

(1) Partial Fraction Expansion

Case 2: Complex Conjugate Poles

Distinct complex conjugate poles (real and imaginary parts affect time-domain behavior)
Treat conjugate pairs together to avoid complex arithmetic

$$\frac{1}{(s-p)(s-p^*)} = \frac{1}{s^2 - (\textcolor{red}{p} + \textcolor{red}{p}^*)s + \textcolor{red}{p}p^*}$$

Case 3: Repeated Poles

$p_1, p_2, p_3 \dots p_n$ multiple identical roots

$$\frac{1}{(s-p_1)^2 (s-p_3)} = \frac{\textcolor{red}{k}_1}{s-p_1} + \frac{k_2}{(s-p_1)^2} + \frac{k_3}{s-p_3}$$

5. Inverse Laplace Transform

Example 1 First-Order Real Poles

$$F(s) = \frac{s+1}{s^2+5s+6}$$

Step 1: Find poles

$$F(s) = \frac{s+1}{(s+2)(s+3)}$$

Step 2: Partial fraction expansion

$$F(s) = \frac{k_1}{s+2} + \frac{k_2}{s+3}$$

Step 3: Determine coefficients

$$F(s) = \frac{-1}{s+2} + \frac{2}{s+3}$$

Step 4: Inverse transform

Using the Laplace pair $\mathcal{L}\left[\mathbf{e^{-\alpha t} u(t)}\right] = \frac{\mathbf{1}}{\mathbf{s + \alpha}}$

$$f(t) = \left(-e^{-2t} + 2e^{-3t}\right)\mathbf{u(t)}$$

5. Inverse Laplace Transform

How to Determine Coefficients k_1, k_2 ?

$$F(s) = \frac{s+1}{(s+2)(s+3)} = \frac{k_1}{s+2} + \frac{k_2}{s+3}$$

Heaviside's "Cover-up" Method

Step 1: Find k_1 , multiply both sides by $s+2$ (to isolate k_1)

$$\text{Right Side} = k_1 + \frac{k_2}{s+3}(s+2)$$

Let $s = -2$, right side = k_1

$$\begin{aligned} \text{Left Side} &= (s+2)F(s)\big|_{s=-2} \\ &= (s+2)\frac{s+1}{(s+2)(s+3)}\bigg|_{s=-2} = -1 \end{aligned}$$

Similarly $k_2 = (s+3)F(s)\big|_{s=-3} = 2$

5. Inverse Laplace Transform

Example 2 First-Order Complex Conjugate Poles

Find the inverse Laplace transform of $F(s) = \frac{s^2 + 3}{(s + 2)(s^2 + 2s + 5)}$

(1) Identify Poles (preserve conjugate terms):

$$F(s) = \frac{s^2 + 3}{(s + 2)\left[(s + 1)^2 + 4\right]}$$

$$\begin{cases} \mathcal{L}\left[e^{-\alpha t} \cos(\omega_0 t) u(t)\right] = \frac{s + \alpha}{(s + \alpha)^2 + \omega_0^2} \\ \mathcal{L}\left[e^{-\alpha t} \sin(\omega_0 t) u(t)\right] = \frac{\omega_0}{(s + \alpha)^2 + \omega_0^2} \end{cases}$$

(2) Partial Fraction Expansion:

Key Laplace Pairs

$$F(s) = k_1 \frac{1}{s + 2} + k_2 \frac{s + 1}{(s + 1)^2 + 2^2} + k_3 \frac{2}{(s + 1)^2 + 2^2}$$

Example 2 First-Order Complex Conjugate Poles

$$\begin{aligned} F(s) &= \frac{s^2 + 3}{(s + 2)(s^2 + 2s + 5)} \\ &= \frac{k_1}{s + 2} + \frac{k_2(s + 1)}{(s + 1)^2 + 4} + \frac{k_3 2}{(s + 1)^2 + 4} \end{aligned}$$

(3) Coefficient Calculation

$$k_1 = (s + 2)F(s) \Big|_{s=-2} = \frac{7}{5}$$

$$F(s) = \frac{7}{5} \frac{1}{s + 2} + \frac{k_2(s + 1)}{(s + 1)^2 + 4} + \frac{k_3 2}{(s + 1)^2 + 4}$$

Combine right-hand side terms and equate numerators

Example 2 First-Order Complex Conjugate Poles

$$F(s) = \frac{\frac{7}{5}(s^2 + 2s + 5) + k_2(s+1)(s+2) + k_3 2(s+2)}{(s+2)(s^2 + 2s + 5)}$$

$$\begin{cases} s^2, \text{ coefficient: } 1 = \frac{7}{5} + k_2 \\ s, \text{ coefficient: } 0 = \frac{14}{5} + 3k_2 + 2k_3 \end{cases} \quad \text{thus} \quad \begin{cases} k_2 = -\frac{2}{5} \\ k_3 = -\frac{4}{5} \end{cases}$$

(4) Inverse transform

$$f(t) = \frac{7}{5} e^{-2t} u(t) + e^{-t} \left[-\frac{2}{5} \cos(2t) - \frac{4}{5} \sin(2t) \right] u(t)$$

Example 3 Multiple-Order Poles

$$F(s) = \frac{s^2}{(s+2)(s+1)^2} = \frac{k_1}{s+2} + \frac{k_2}{s+1} + \frac{k_3}{(s+1)^2}$$

k_1 Coefficient for simple pole, k_3 Highest-order coefficient for repeated pole

$$k_1 = (s+2) \frac{s^2}{(s+2)(s+1)^2} \Big|_{s=-2} = 4$$

$$k_3 = (s+1)^2 \frac{s^2}{(s+2)(s+1)^2} \Big|_{s=-1} = 1$$

How to determine k_2 ?

5. Inverse Laplace Transform

Example 3 Multiple-Order Poles, Solving for k_2

$$F(s) = \frac{s^2}{(s+2)(s+1)^2} = \frac{k_1}{s+2} + \frac{k_2}{s+1} + \frac{k_3}{(s+1)^2}$$

Isolate k_2 by eliminating other terms

Multiply by $(s+1)^2$

$$\text{Right} = (s+1)^2 \frac{k_1}{s+2} + k_2(s+1) + k_3$$

Differentiate Both Sides with respect to s

$$\text{Right} = \left[(s+1)^2 \left(\frac{k_1}{s+2} \right)' + 2(s+1) \frac{k_1}{s+2} \right] + k_2 + 0$$

Evaluate at $s = -1$, $\text{Right} = k_2$

$$k_2 = \text{left} = \frac{d}{ds} \left[\frac{s^2}{s+2} \right]_{s=-1} = \frac{s^2 + 4s}{(s+2)^2} \Big|_{s=-1} = -3$$

5. Inverse Laplace Transform

Example 3 Multiple-Order Poles

$$F(s) = \frac{4}{s+2} + \frac{-3}{s+1} + \frac{1}{(s+1)^2}$$

$$\frac{1}{s^{n+1}} \rightarrow \frac{1}{n!} t^n u(t)$$

$$\frac{1}{(s+\alpha)^{n+1}} \rightarrow e^{-\alpha t} \frac{1}{n!} t^n u(t)$$

$$f(t) = \mathcal{L}^{-1}[F(s)] = \left(4e^{-2t} - 3e^{-t} + te^{-t} \right) u(t)$$

Summary

Partial Fraction Expansion

$$F(s) = \frac{s+1}{s^2+5s+6}$$

$$F(s) = \frac{s+\gamma}{(s+\alpha)^2+\beta^2}$$

$$F(s) = \frac{s^2}{(s+2)(s+1)^2}$$

- Identify Poles of $F(s)$
- Expand $F(s)$ into Partial Fractions
- Inverse Transform $f(t)$ via Laplace Table

Main Content

- Introduction to Analyzing Dynamic Circuits Using Laplace Transform Method
- s-Domain Equivalent Models of Circuits
- Dynamic Circuit Analysis

1. Introduction to Analyzing Dynamic Circuits

- ✓ First, establish the s-domain equation (two approaches)
 - Formulate the time-domain differential equation, then apply the differentiation and integration properties to obtain the Laplace transform (**this method is not used here**)
 - **Directly construct the algebraic equation based on the s-domain model of the circuit**
- ✓ Solve the s-domain equation

$$F(s) \rightarrow f(t)$$

to obtain the time-domain solution

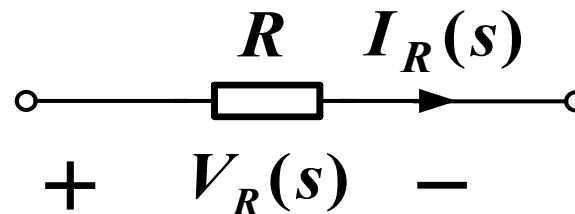
2. s-Domain Equivalent Models of Circuits

Circuit **components** are represented by their **complex frequency-domain** models, and **excitations** are replaced by their **Laplace** transforms, thereby forming the **s-domain model** of the circuit

(1) s-Domain Model of a Resistor

Time-domain relationship: $v_R(t) = Ri_R(t)$

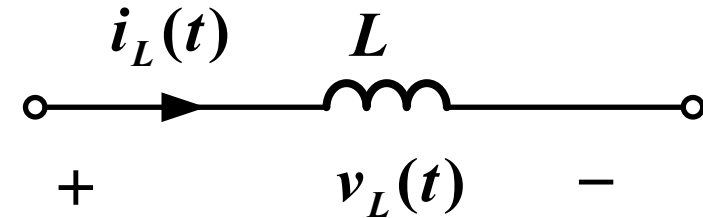
Transform-domain relationship: $V_R(s) = RI_R(s)$



6. LT Analysis of Dynamic Circuits

(2) s-Domain Model of an Inductor

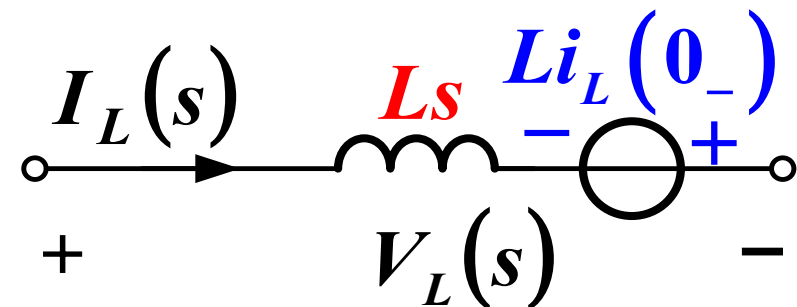
Time-domain
relationship: $v_L(t) = L \frac{di_L(t)}{dt}$



$$\text{Let } \begin{cases} \mathcal{L}[i_L(t)] = I_L(s) \\ \mathcal{L}[v_L(t)] = V_L(s) \end{cases}$$

Applying the time-domain differentiation property of the Laplace transform:

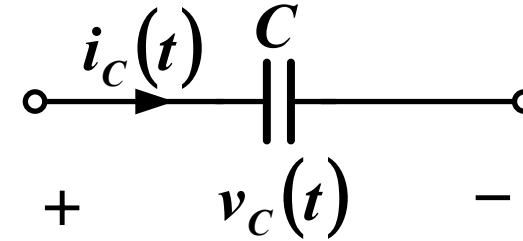
$$\begin{aligned} V_L(s) &= L[sI_L(s) - i_L(0_-)] \\ &= LsI_L(s) - Li_L(0_-) \end{aligned}$$



(3) s-Domain Model of a Capacitor

Time-domain relationship: $v_C(t) = \frac{1}{C} \int_{-\infty}^t i_C(\tau) d\tau$

Let $\begin{cases} \mathcal{L}[i_C(t)] = I_C(s) \\ \mathcal{L}[v_C(t)] = V_C(s) \end{cases}$



Using the integration property of the unilateral Laplace transform:

$$V_C(s) = \frac{1}{C} \left[\frac{\textcolor{red}{I}_C(s)}{\textcolor{red}{s}} + \frac{i_C^{(-1)}(0_-)}{s} \right]$$

s-Domain relationship:

$$V_C(s) = \frac{1}{sC} \textcolor{red}{I}_C(s) + \frac{\textcolor{blue}{v}_C(0_-)}{s}$$

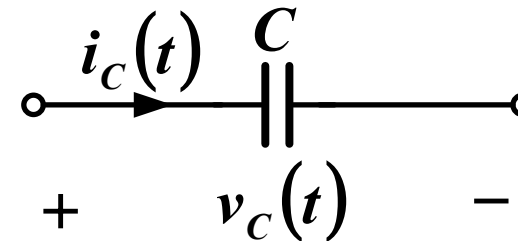
$$\begin{aligned} \frac{1}{C} i_C^{(-1)}(0_-) &= \frac{1}{C} \int_{-\infty}^{0_-} i_C(\tau) d\tau \\ &= v_C(0_-) \end{aligned}$$

6. LT Analysis of Dynamic Circuits

(3) s-Domain Model of a Capacitor

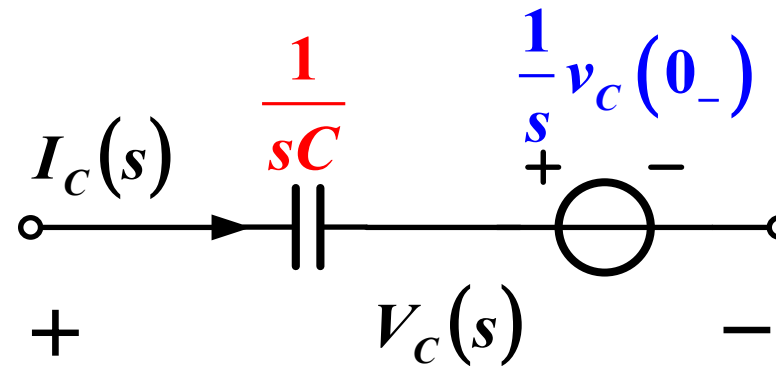
Time-domain relationship:

$$v_C(t) = \frac{1}{C} \int_{-\infty}^t i_C(\tau) d\tau$$



s-Domain relationship:

$$V_C(s) = \frac{1}{sC} I_C(s) + \frac{v_C(0_-)}{s}$$



3. Dynamic Circuit Analysis

(1) Complex Frequency-Domain Forms of Circuit Theorems

$$i(t) \xleftrightarrow{LT} I(s), \quad \text{KCL} : \sum i(t) = 0 \rightarrow \sum I(s) = 0$$

$$v(t) \xleftrightarrow{LT} V(s) \quad \text{KVL} : \sum v(t) = 0 \rightarrow \sum V(s) = 0$$

All standard linear circuit analysis techniques remain valid in the s-domain

3. Dynamic Circuit Analysis

(2) Steps for Solving Responses

- Draw the $t=0$ equivalent circuit to determine the initial values of state variables $v_C(0_-)$ and $i_L(0_-)$
- Construct the s-domain equivalent model
- Formulate s-domain equations (algebraic equations)
- Solve the s-domain equations to obtain the Laplace transform of the response $V(s)$ or $I(s)$
- Perform the inverse Laplace transform to find $v(t)$ or $i(t)$

6. LT Analysis of Dynamic Circuits

Example 2

The circuit shown below has zero initial state. At $t = 0$, switch S closes, connecting to DC power source E . Find the current $i(t)$

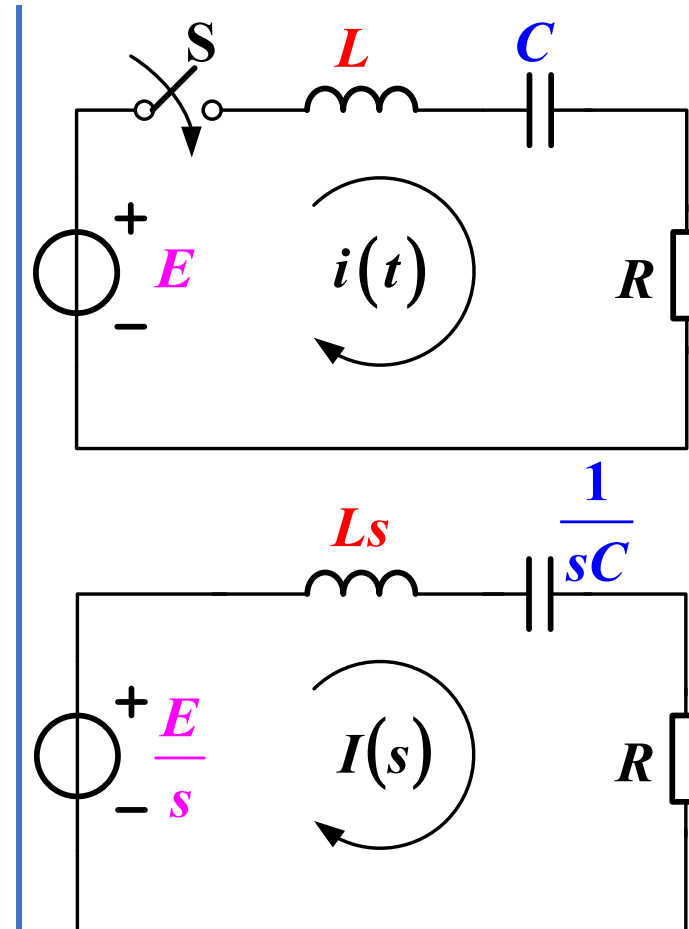
(1) Zero initial conditions:

$$\Rightarrow i_L(0_-) = 0\text{A}, v_C(0_-) = 0\text{V}$$

(2) Draw the s-domain equivalent model for $t > 0$

(3) Write KVL equation:

$$\frac{E}{s} = LsI(s) + \frac{1}{Cs}I(s) + RI(s)$$



6. LT Analysis of Dynamic Circuits

Example 2

$$\frac{E}{s} = LsI(s) + \frac{1}{Cs}I(s) + RI(s)$$

Obtain

$$I(s) = \frac{E}{s\left(Ls + R + \frac{1}{sC}\right)} = \frac{E}{L} \frac{1}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

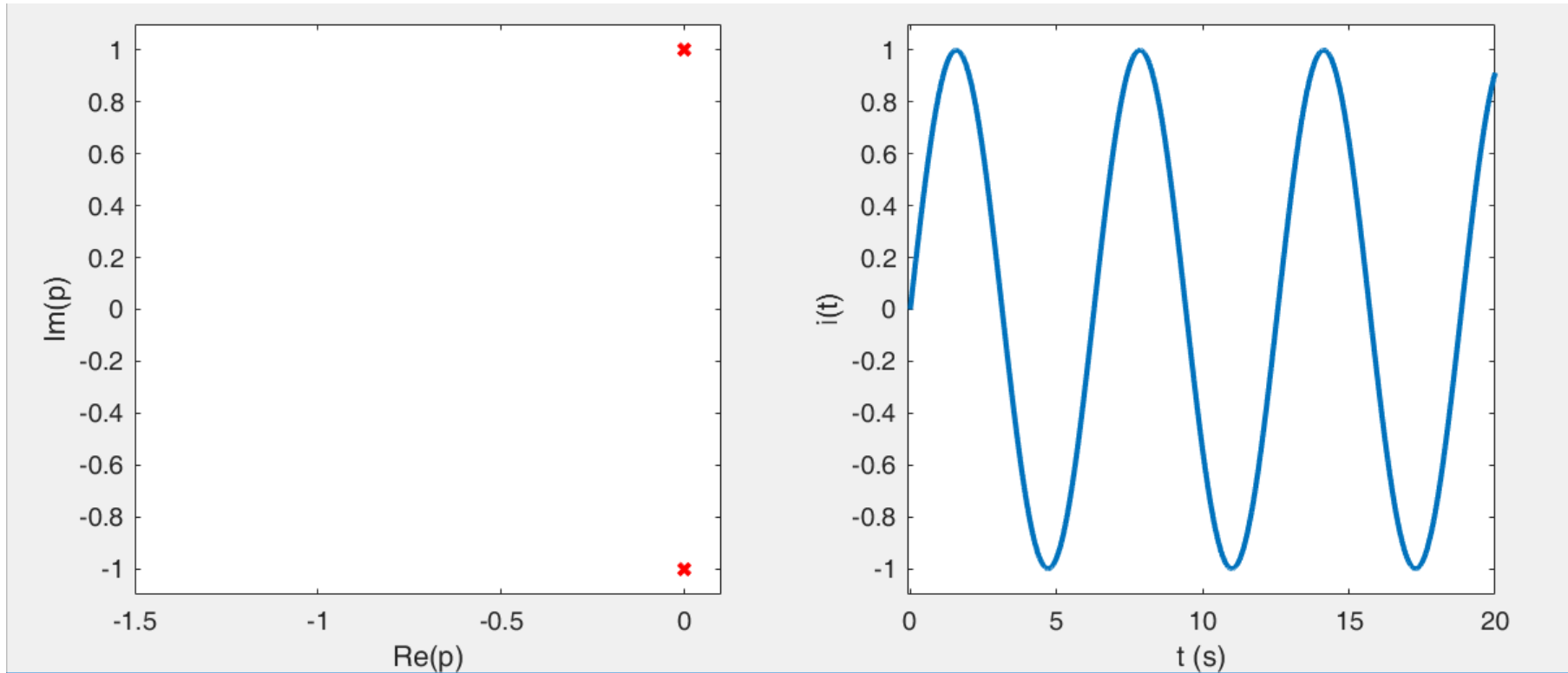
Poles can be expressed as

$$p_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$p_2 = -\frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

6. LT Analysis of Dynamic Circuits

Example 2

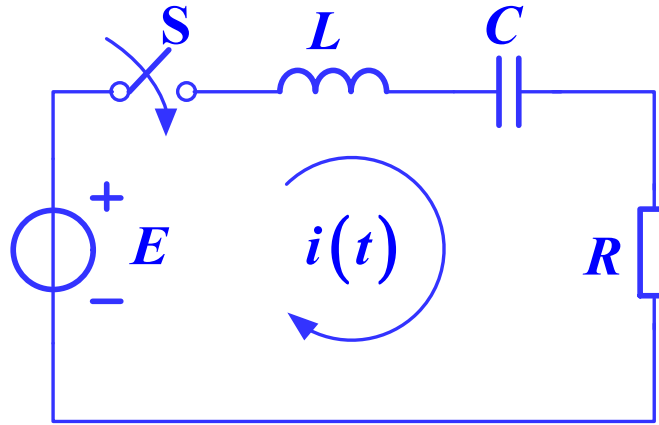


$$p_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$p_2 = -\frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

6. LT Analysis of Dynamic Circuits

Example 2

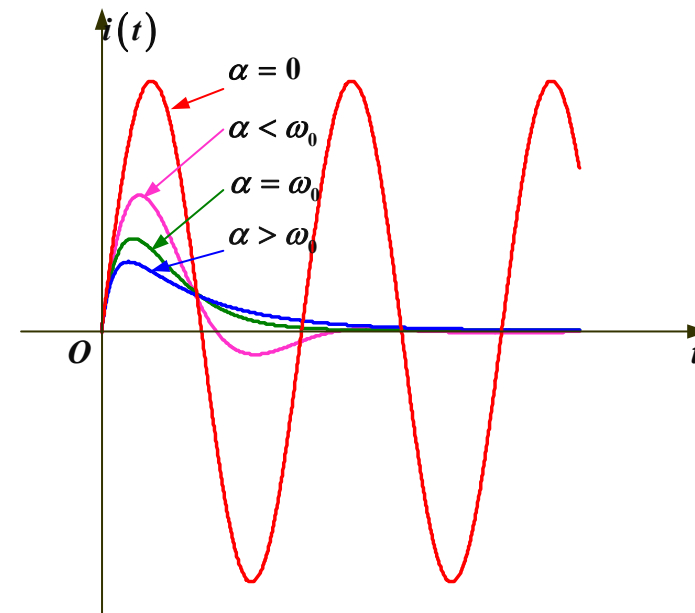


$$i(t) = \frac{E}{L(p_1 - p_2)} (e^{p_1 t} - e^{p_2 t}) u(t)$$

$$\text{let } \alpha = \frac{R}{2L}, \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\text{Then } p_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}, \quad p_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

Real part Imaginary part



1. **Undamped** oscillation: constant amplitude oscillation
2. **Underdamped** oscillation: Exponentially decaying oscillation
3. **Critical damping**: Fastest response without overshoot
4. **Overdamping**: Slow non-oscillatory response

Summary

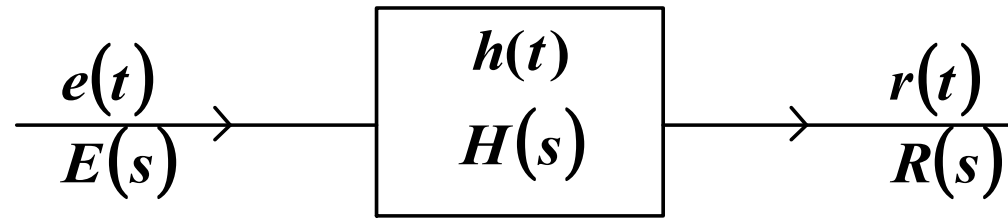
Analyzing Dynamic Circuits Using Laplace Transform Method

- ✓ **Directly construct the algebraic equation based on the s-domain model of the circuit**
- ✓ **Solve the s-domain equation**
- ✓ **$F(s) \rightarrow f(t)$, to obtain the time-domain solution**

Mian Content

- **Definition of System Function**
- **Transformation of Continuous System Models**
- **Interconnection of Continuous Systems**

1. Definition of System Function



Time-domain analysis $r(t) = e(t) * h(t)$

Let $E(s) = \mathcal{L}[e(t)]$, $H(s) = \mathcal{L}[h(t)]$, then

$$\mathcal{L}[r(t)] = R(s) = E(s) \cdot H(s)$$

Definition: system function $H(s) = \frac{\text{Laplace transform of the zero-state response}}{\text{Laplace transform of the input signal}}$

$$H(s) = \frac{R(s)}{E(s)}$$

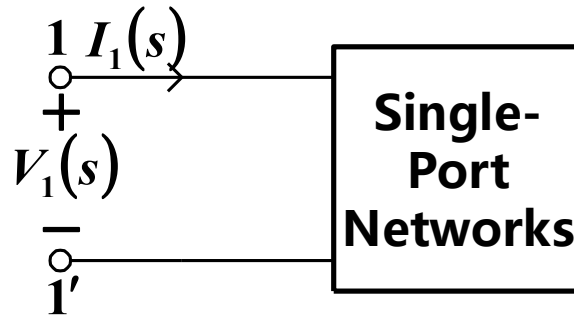
2. Transformation of Continuous System Models

The system function serves as a bridge for converting between different representations of continuous systems:

- **Circuit $\rightarrow$ s -Domain Equivalent Model**
- **Differential Equation $\rightarrow$ System Function $H(s)$**
- **Interconnected Systems**

(1) Determining System Functions from Circuit Diagrams

Driving-Point Functions: excitation and response are measured at the same port

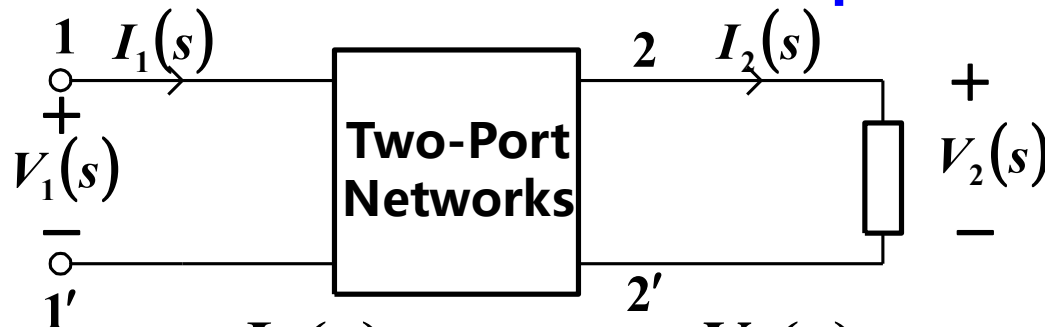


$$H(s) = \frac{I_1(s)}{V_1(s)}$$

$$H(s) = \frac{V_1(s)}{I_1(s)}$$

Driving-Point Admittance Driving-Point Impedance

Transfer Functions: excitation and response are measured at different ports



$$H(s) = \frac{I_2(s)}{V_1(s)}$$

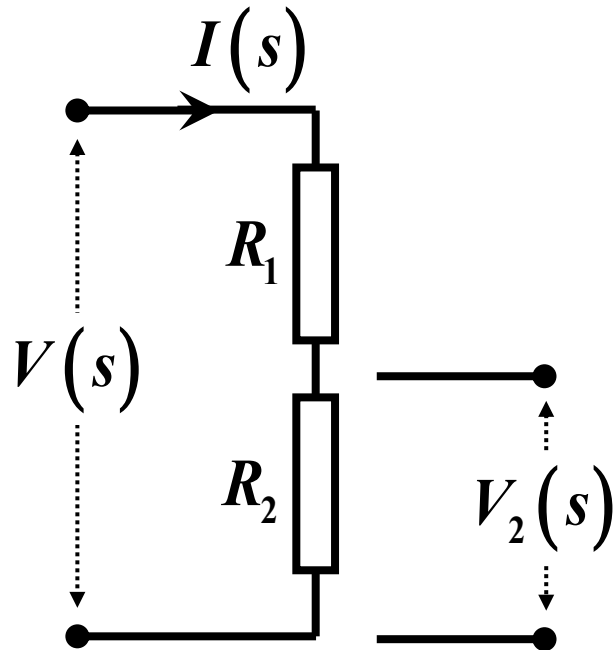
$$H(s) = \frac{V_2(s)}{I_1(s)}$$

$$H(s) = \frac{V_2(s)}{V_1(s)}$$

$$H(s) = \frac{I_2(s)}{I_1(s)}$$

Transfer Admittance Transfer Impedance Voltage Transfer Ratio Current Transfer Ratio

Example 1



(1) Driving-Point Admittance

Excitation: Voltage $V(s)$

Response: Current $I(s)$

The driving-point **admittance** is defined as:

$$H_1(s) = \frac{I(s)}{V(s)} = \frac{1}{R_1 + R_2}$$

(2) Voltage Transfer Function

Excitation: Voltage $V(s)$

Response: Voltage $V_2(s)$ across R_2

The **voltage transfer function** is:

$$H_2(s) = \frac{V_2(s)}{V(s)} = \frac{R_2}{R_1 + R_2}$$

(2) Deriving System Functions from Differential Equations

$$\sum_{k=0}^n a_k \frac{d^k r(t)}{dt^k} = \sum_{j=0}^m b_j \frac{d^j e(t)}{dt^j}$$

Taking the **bilateral** Laplace transform of both sides and applying the time-domain differentiation property of Laplace transforms yields

$$\sum_{k=0}^n a_k s^k R(s) = \sum_{j=0}^m b_j s^j E(s)$$

System Function Derivation

$$H(s) = \frac{R(s)}{E(s)} = \frac{\sum_{j=0}^m b_j s^j}{\sum_{k=0}^n a_k s^k}$$

Example 2: Solving for System Function and Unit Impulse Response of a Causal System

$$\frac{d^2 r(t)}{dt^2} + 5 \frac{dr(t)}{dt} + 6r(t) = \frac{d^2 e(t)}{dt^2} + \frac{de(t)}{dt}$$

Apply the **bilateral Laplace transform** to both sides of the equation (assuming zero initial conditions)

$$s^2 R(s) + 5sR(s) + 6R(s) = s^2 E(s) + sE(s)$$

$$(s^2 + 5s + 6)R(s) = (s^2 + s)E(s)$$

$$H(s) = \frac{R(s)}{E(s)} = \frac{s^2 + s}{s^2 + 5s + 6}$$

Example 2

$$H(s) = \frac{s^2 + s}{s^2 + 5s + 6} = 1 - \frac{4s + 6}{s^2 + 5s + 6}$$

$$\text{let } \frac{4s + 6}{s^2 + 5s + 6} = \frac{k_1}{s + 2} + \frac{k_2}{s + 3}$$

$$k_1 = (s + 2) \frac{4s + 6}{s^2 + 5s + 6} \Big|_{s=-2} = -2$$

$$k_2 = (s + 3) \frac{4s + 6}{s^2 + 5s + 6} \Big|_{s=-3} = 6$$

$$H(s) = 1 + \frac{2}{s + 2} - \frac{6}{s + 3}$$

$$\text{Thus } h(t) = \delta(t) + 2e^{-2t}u(t) - 6e^{-3t}u(t)$$

(3) Zeros and Poles of System Functions

Factored form

$$H(s) = \frac{B(s)}{A(s)} = K \frac{(s - z_1)(s - z_2) \cdots (s - z_j) \cdots (s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_k) \cdots (s - p_n)}$$

$$= K \frac{\prod_{j=1}^m (s - z_j)}{\prod_{k=1}^n (s - p_k)}$$

Zeros $z_1, z_2 \cdots z_n$ make $H(s)=0$

Poles $p_1, p_2 \cdots p_n$ make $H(s)$ unbounded

K: Scalar multiplier

Zero-Pole Plot in the s-Plane:

Poles: Represented by $\times$ (cross)

Zeros: Represented by $\bigcirc$ (circle)

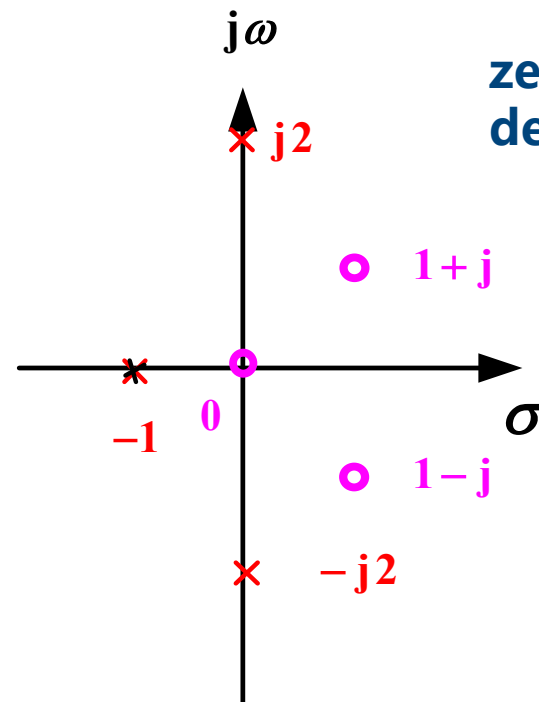
Example 3

$$H(s) = \frac{s(s-1+j1)(s-1-j1)}{(s+1)^2(s+j2)(s-j2)}$$

Poles: $p_1 = p_2 = -1$, $p_3 = -j2$, $p_4 = j2$

Zeros: $z_1 = 0$, $z_2 = 1-j1$, $z_3 = 1+j1$, $z_4 = \infty$

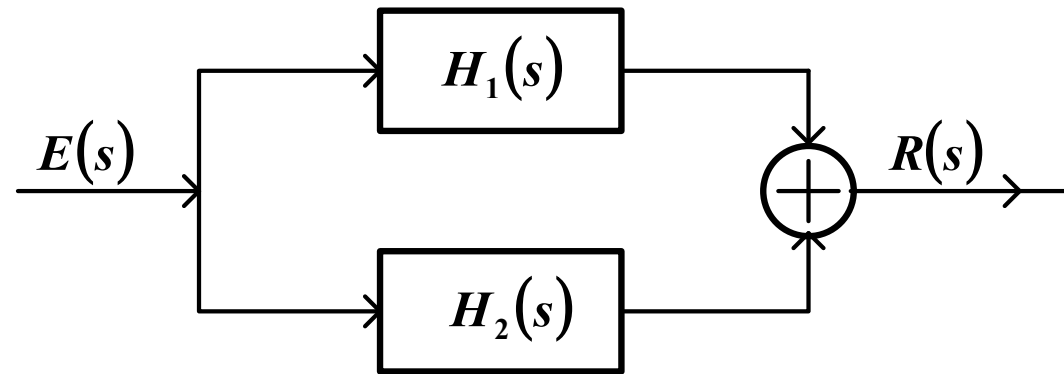
Zero-Pole Plot:



zero at infinity, due to numerator degree < denominator

3. Interconnected Systems

(1) Parallel Connection of LTI Systems

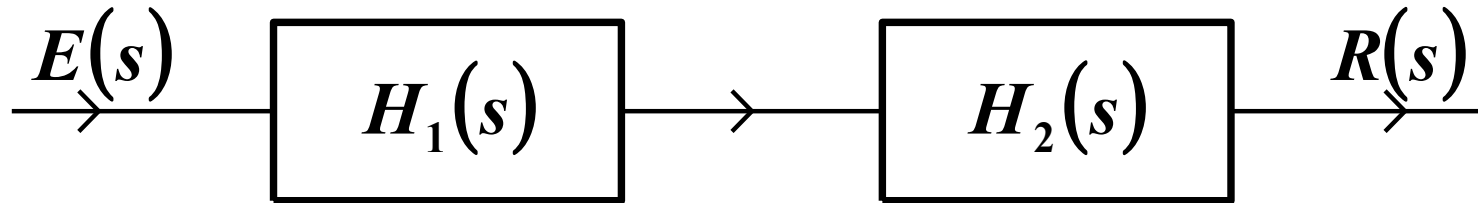


$$\begin{aligned} R(s) &= E(s) H_1(s) + E(s) H_2(s) \\ &= E(s) H(s) \end{aligned}$$

$$H(s) = H_1(s) + H_2(s)$$

3. Interconnected Systems

(2) Cascade Connection of LTI Systems

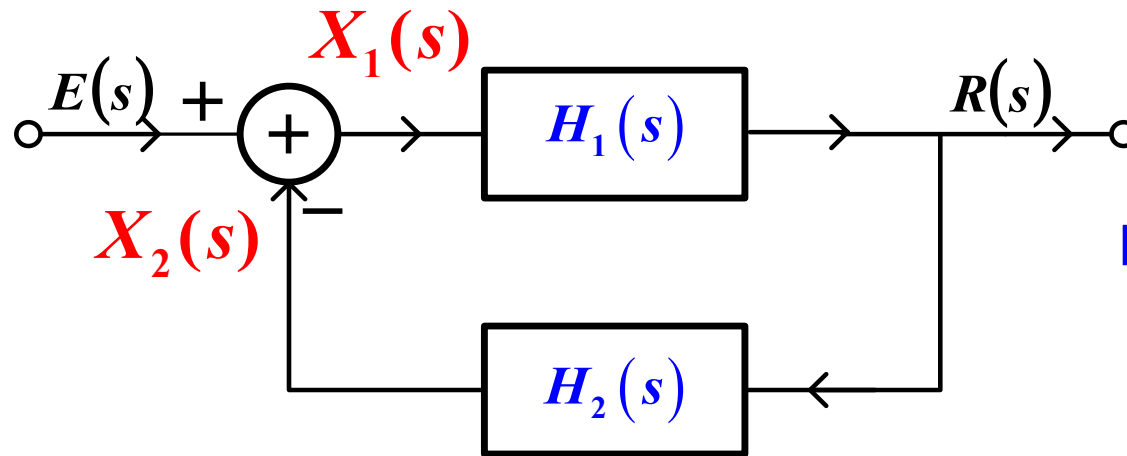


$$\begin{aligned} R(s) &= E(s) H_1(s) \cdot H_2(s) \\ &= E(s) \cdot H(s) \end{aligned}$$

$$H(s) = H_1(s) \cdot H_2(s)$$

3. Interconnected Systems

(3) LTI Feedback Systems



Adder Equation

$$X_1(s) = E(s) - X_2(s)$$

Forward Path

$$R(s) = X_1(s)H_1(s)$$

Feedback Path

$$X_2(s) = R(s) \cdot H_2(s)$$

3. Interconnected Systems

(3) LTI Feedback Systems

$$X_1(s) = E(s) - X_2(s)$$

$$R(s) = X_1(s)H_1(s)$$

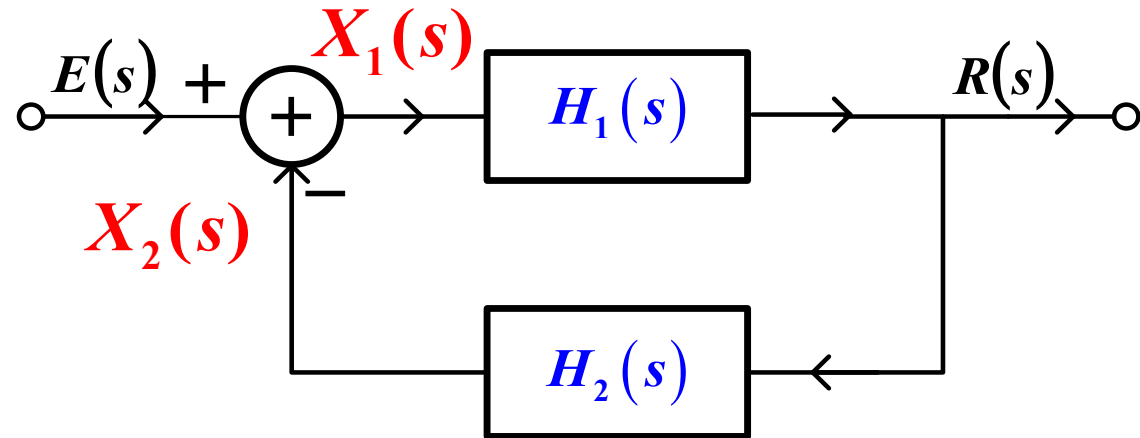
$$X_2(s) = R(s) \cdot H_2(s)$$

Substitution

$$\frac{R(s)}{H_1(s)} = E(s) - R(s) \cdot H_2(s)$$

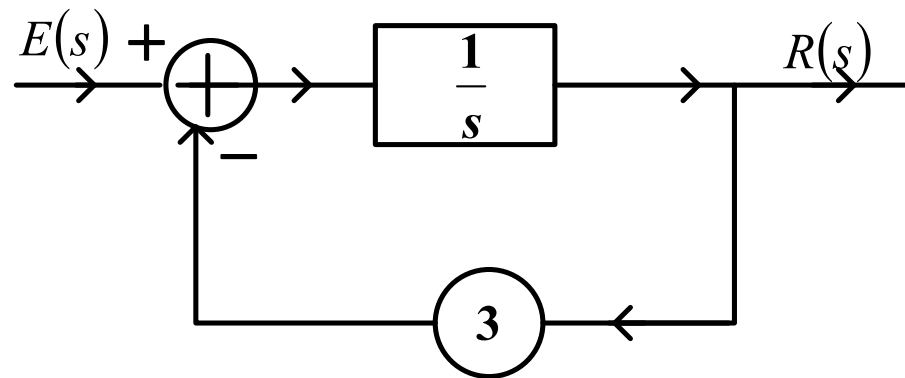
Closed-loop transfer function

$$H(s) = \frac{R(s)}{E(s)} = \frac{H_1(s)}{1 + H_1(s)H_2(s)}$$



Example 4

Given the system block diagram below, derive the system function and the differential equation describing this system



$$H_1(s) = \frac{1}{s} \quad H_2(s) = 3$$

$$H(s) = \frac{R(s)}{E(s)} = \frac{H_1(s)}{1 + H_1(s)H_2(s)} = \frac{1}{s + 3}$$

Example 4

$$H(s) = \frac{R(s)}{E(s)} = \frac{1}{s+3}$$

s-Domain Equation: the relationship between excitation and response

$$sR(s) + 3R(s) = E(s)$$

Time-Domain Differential Equation: Apply the inverse Laplace transform

$$\frac{dr(t)}{dt} + 3r(t) = e(t)$$

Summary

- Definition of System Function
- Transformation of Continuous System Models
- Interconnection of Continuous Systems

$$H(s) = \frac{R(s)}{E(s)}$$

Main Content

- **Assessing System Stability via Impulse Response**
- **Determining System Stability from Zero-Pole Distribution of the System Function**

1. Problem Statement

Given the transfer function of a continuous-time system:

$$H(s) = \frac{1}{s+1} + \frac{0.001}{s-2}$$

When the input is a unit step signal $u(t)$, the Laplace transform of the zero-state response is:

$$\begin{aligned} R_{zs}(s) &= \frac{H(s)}{s} \\ &= \frac{1 - 0.0005}{s} - \frac{1}{s+1} + \frac{0.0005}{s-2} \end{aligned}$$

$$r_{zs}(t) = \left(1 - e^{-t} + 0.0005 e^{2t}\right) u(t)$$

Though $0.0005 \ll 1...$

System Instability: The presence of a right half-plane (RHP) pole at $s=2$ makes the system unstable

Engineering Insight: Even with negligible coefficients for unstable modes, long-term behavior is governed by the fastest-growing exponential term

2. General Criteria for System Stability

(1) Definition:

A system is said to be **Bounded-Input Bounded-Output (BIBO) stable** (or simply stable) if, for any **bounded input**, its zero-state **response** is also **bounded**

Necessary and Sufficient Condition for Stability (Absolute Integrability Condition):

$$\int_{-\infty}^{\infty} |h(t)| dt \leq M$$

where M is a finite positive constant

2. General Criteria for System Stability

Time-Domain Stability Criterion: $\int_{-\infty}^{\infty} |h(t)| dt \leq M$

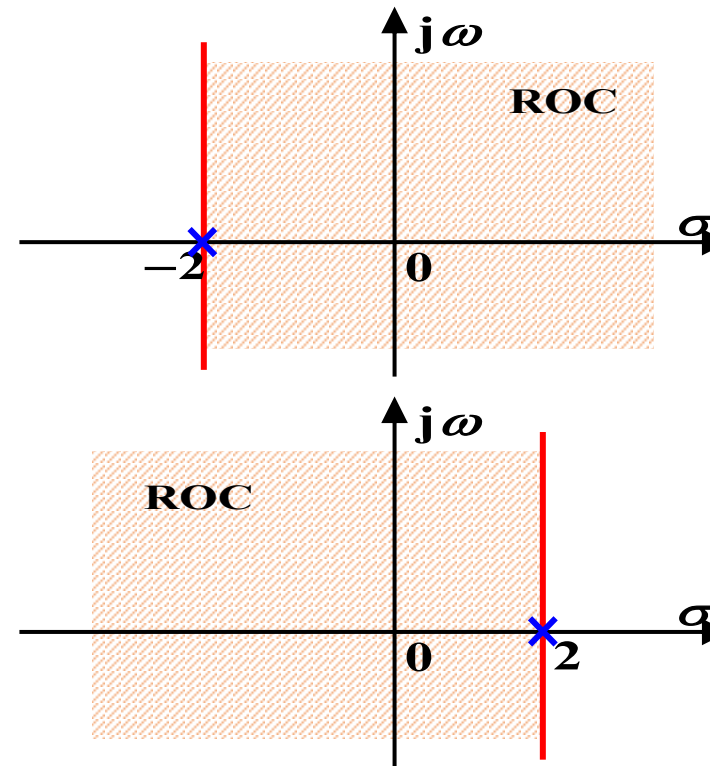
s-Domain Stability Criterion: A system is stable if the **ROC** of its transfer function $H(s) = \mathcal{L}[h(t)]$ **includes the imaginary axis**

$$h(t) = e^{-2t} u(t)$$

$$H(s) = \frac{1}{s+2} \quad (\sigma > -2)$$

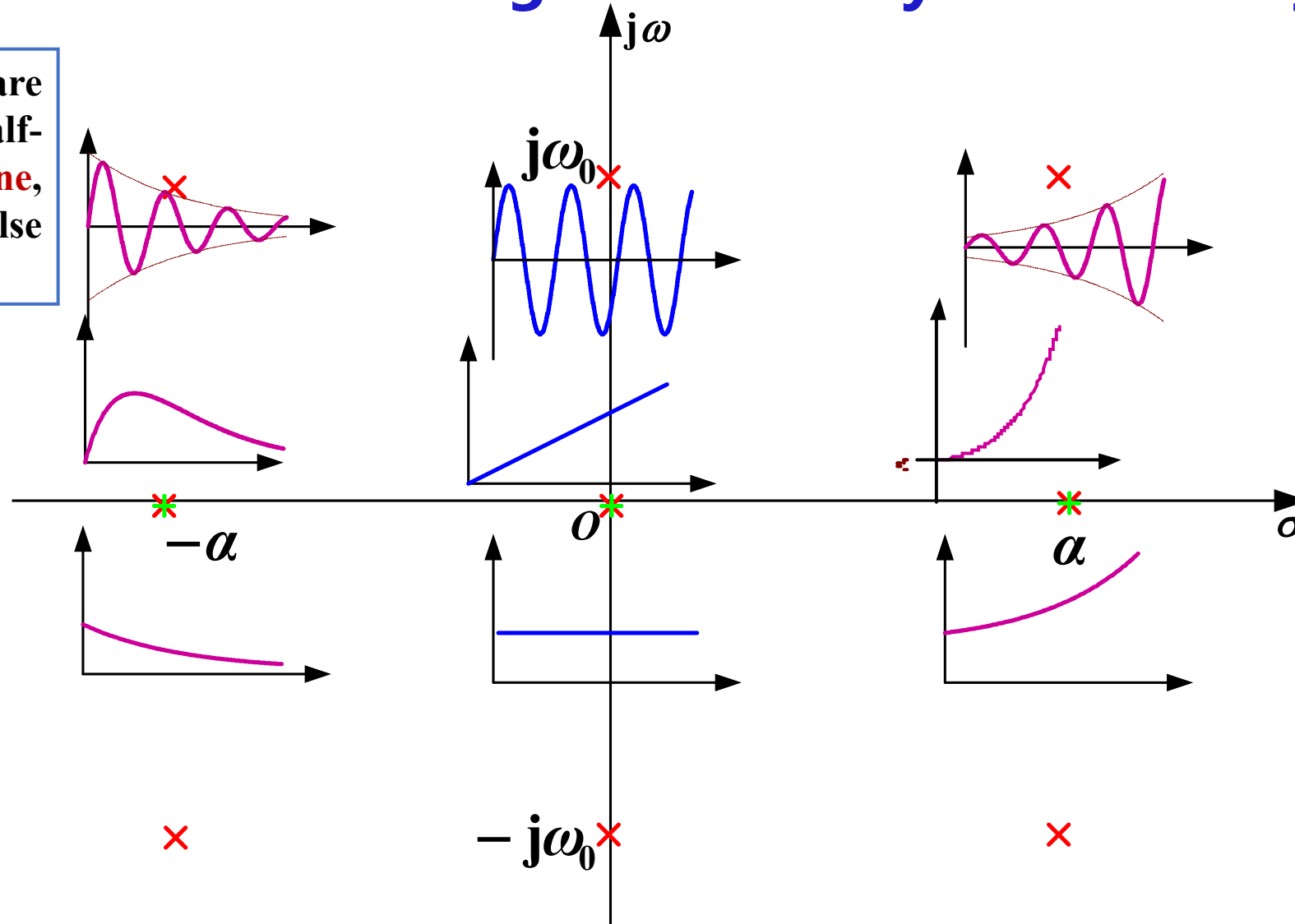
$$h(t) = e^{2t} u(-t)$$

$$H(s) = -\frac{1}{s-2} \quad (\sigma < 2)$$



3. Method for Determining the Stability of Causal Systems

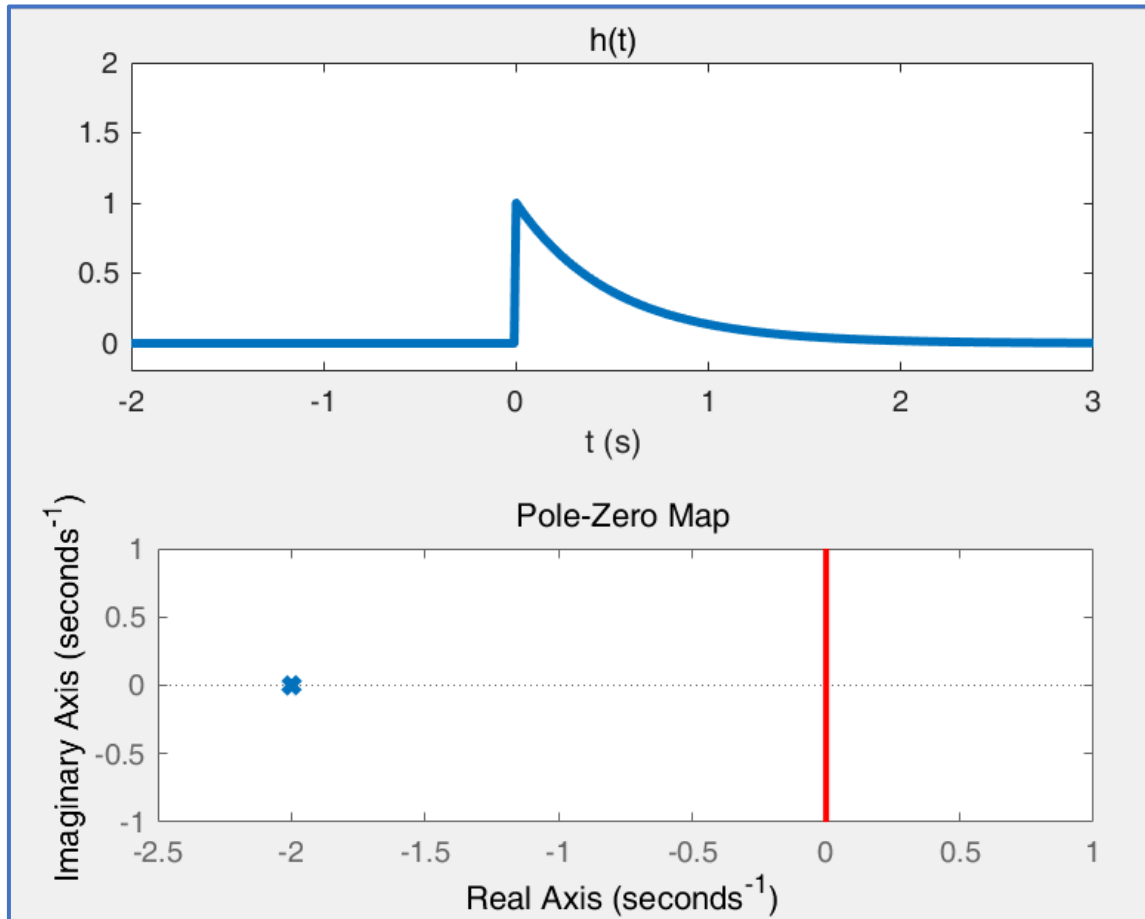
When all poles are located in the **left** half-plane of the **s-plane**, the unit impulse response **converges**



3. Method for Determining the Stability of Causal Systems

Single Real Pole

$$H(s) = \frac{1}{s + \alpha} \quad h(t) = e^{-\alpha t} u(t)$$

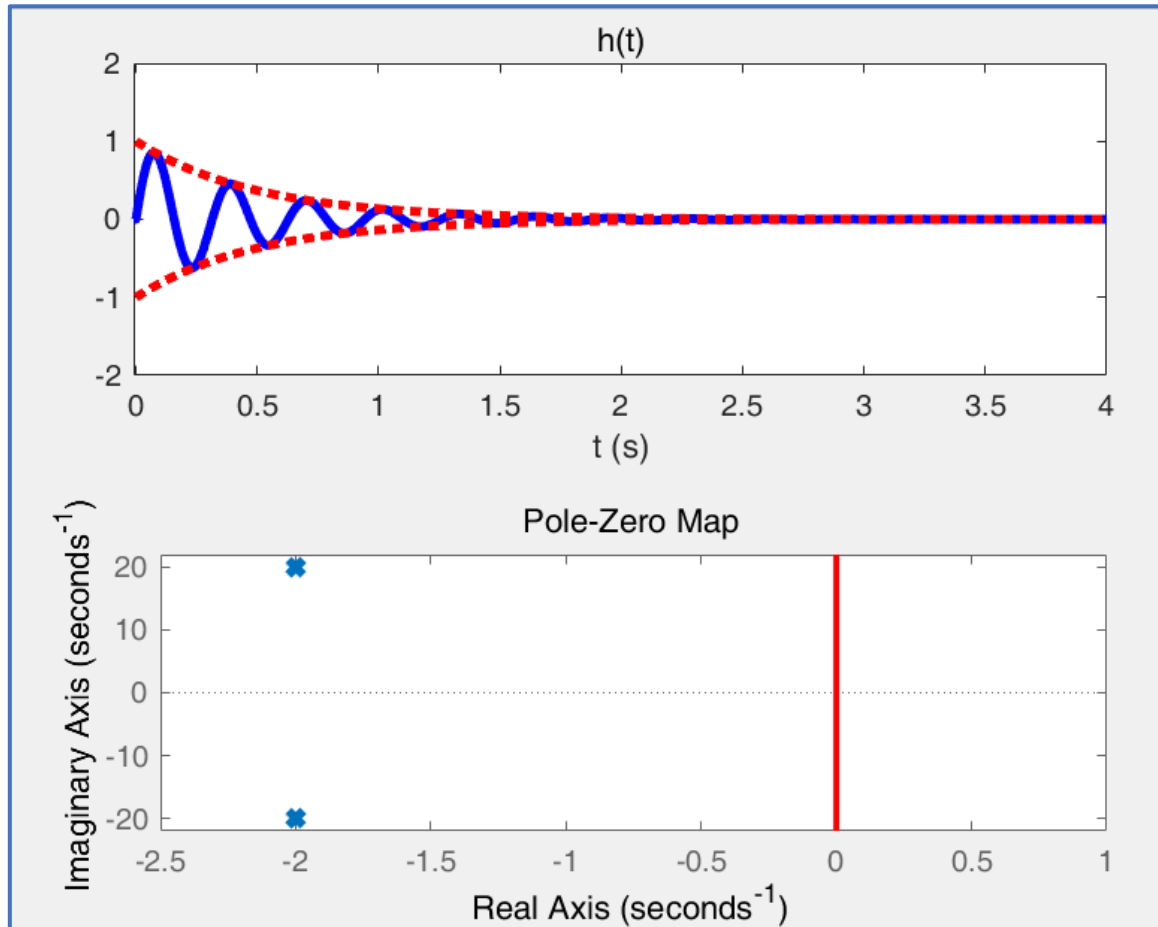


- Pole on the **Left** Real Axis
 $h(t)$ Exponential **decay**
- Pole at the **Origin**
 $h(t)$ **Constant**
- Pole on the **Right** Real Axis
 $h(t)$ Exponential **growth**

3. Method for Determining the Stability of Causal Systems

Conjugated complex poles

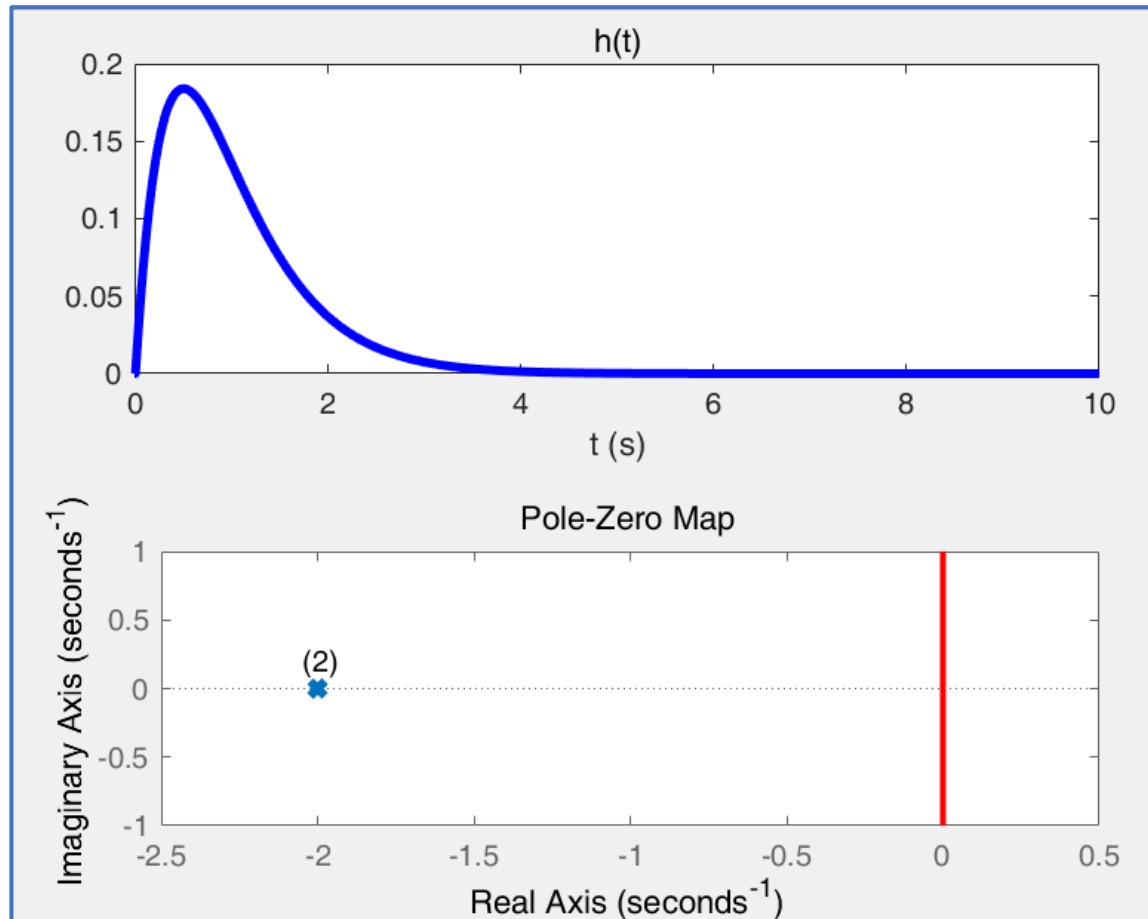
$$H(s) = \frac{\omega_0}{(s + \alpha)^2 + \omega_0^2} \quad h(t) = e^{-\alpha t} \sin(\omega_0 t) \cdot u(t)$$



- Pole in the **Left** Half-Plane
 $h(t)$ **Decaying** oscillation
- Pole on the **Imaginary** Axis
 $h(t)$ **Sustained** oscillation
- Pole in the **Right** Half-Plane
 $h(t)$ **Growing** oscillation

3. Method for Determining the Stability of Causal Systems

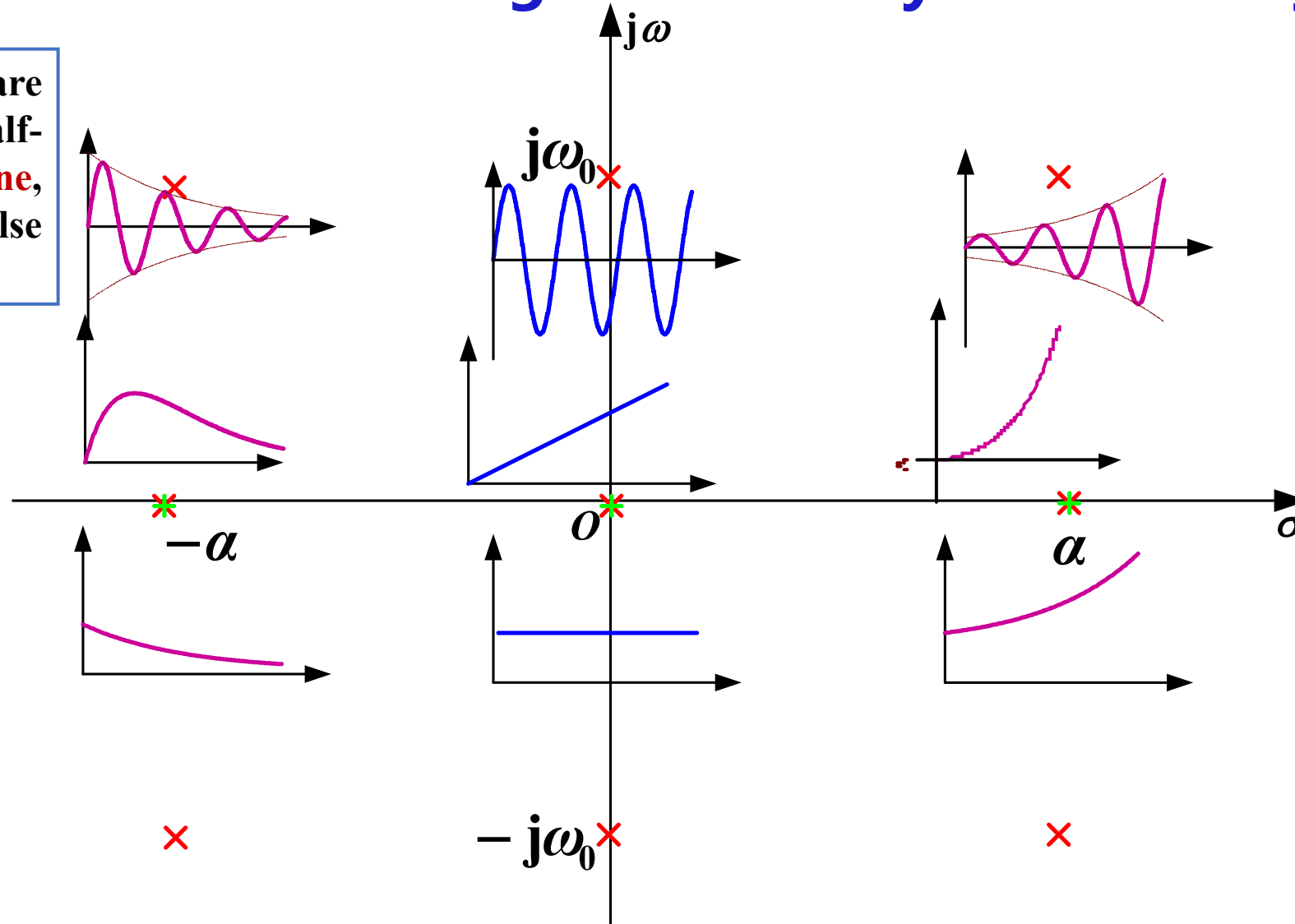
Repeated poles $H(s) = \frac{1}{(s + \alpha)^2}$ $h(t) = t e^{-\alpha t} u(t)$



- Pole on the **Left** Real Axis
 $h(t)$ **decaying** oscillation
- Pole at the **Origin**
 $h(t)$ **linear growth** (marginally unstable)
- Pole on the **Right** Real Axis
 $h(t)$ **Exponential growth**

3. Method for Determining the Stability of Causal Systems

When all poles are located in the **left** half-plane of the **s-plane**, the unit impulse response **converges**



3. Method for Determining the Stability of Causal Systems

(1) Stable System

A system is stable if all poles of $H(s)$ lie strictly in the **left half-plane** (LHP) of the s -plane (*excluding* the imaginary axis). This ensures:

$$\lim_{t \rightarrow \infty} h(t) = 0 \text{ (Impulse response decays to zero)}$$

(2) Unstable System

A system is unstable if: Any pole lies in **the right half-plane** (RHP), or there are repeated poles (second-order or higher) on the imaginary axis,

$$\lim_{t \rightarrow \infty} h(t) \rightarrow \infty \text{ (Impulse response diverges)}$$

(3) Marginally Stable (Technically Unstable) System

A system is marginally stable if: Poles are simple (first-order) and lie exactly on the imaginary axis

$$\text{As } t \rightarrow \infty, h(t) \text{ approaches } a \begin{cases} \text{non-zero constant,} \\ \text{or sustained oscillation.} \end{cases}$$

7.2 Stability of Continuous-Time Systems

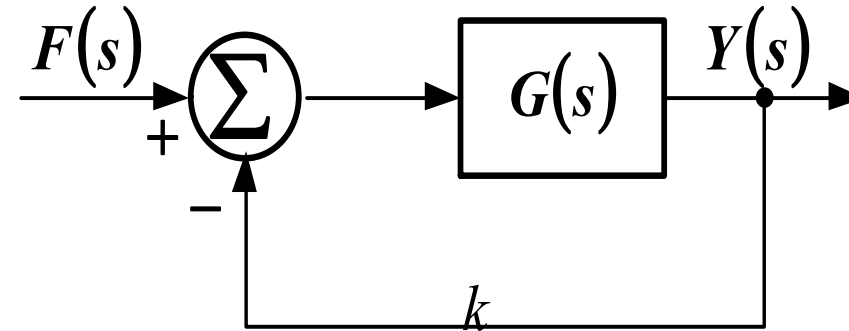
Example 1

A causal feedback system is shown in the figure

The transfer function of the subsystem is

$$G(s) = \frac{1}{(s-1)(s+2)}$$

This subsystem is unstable



Under what condition of the constant k is the system stable?

Based on the given conditions, the equation is derived as:

$$H(s) = \frac{Y(s)}{F(s)} = \frac{G(s)}{1 + kG(s)} = \frac{1}{s^2 + s - 2 + k}$$

7.2 Stability of Continuous-Time Systems

Example 1

$$H(s) = \frac{1}{s^2 + s - 2 + k}$$

The poles of $H(s)$

$$p_{1,2} = -\frac{1}{2} \pm \sqrt{\frac{9}{4} - k}$$

For all poles to lie in the left half-plane of s , the following must hold

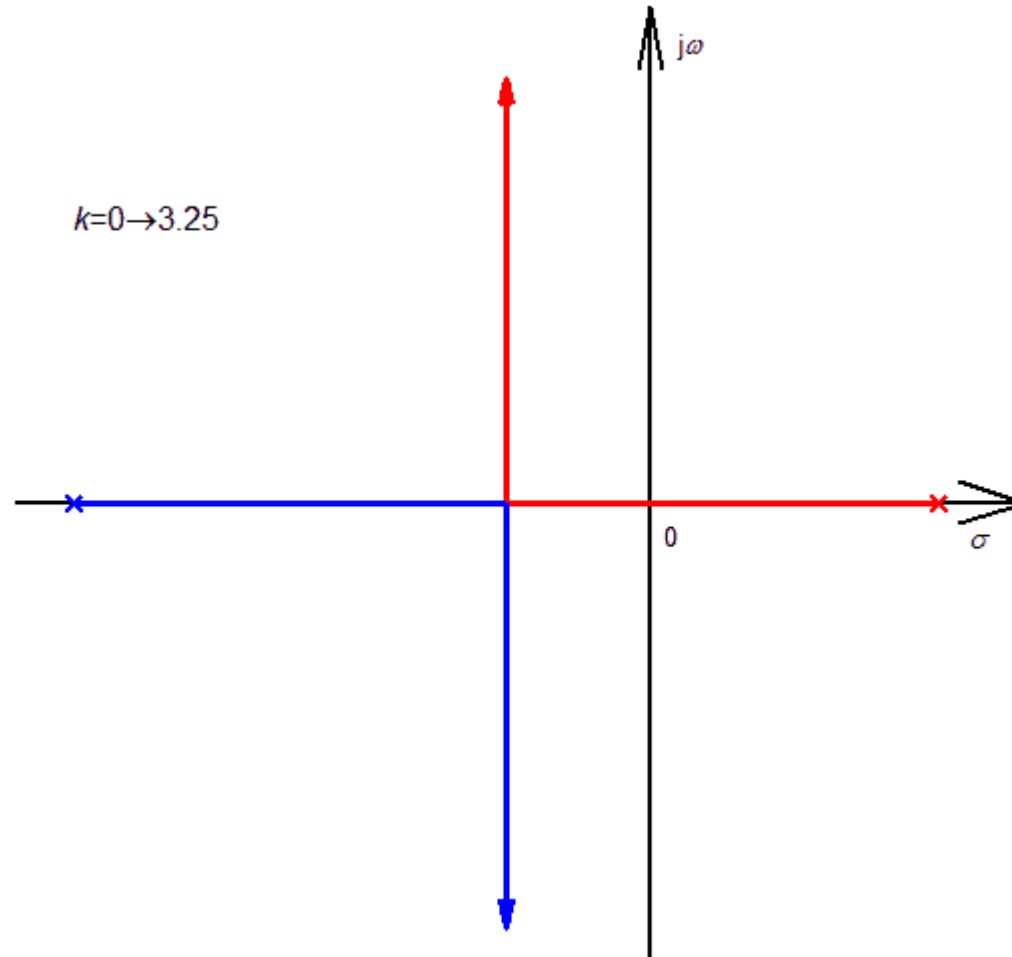
$$\frac{9}{4} - k < 0 \quad \text{or} \quad \begin{cases} \frac{9}{4} - k \geq 0 \\ -\frac{1}{2} + \sqrt{\frac{9}{4} - k} < 0 \end{cases}$$

Solving these conditions yields $k > 2$. Therefore, the system is stable when $k > 2$

7.2 Stability of Continuous-Time Systems

Example 1

By introducing **negative feedback**, the **unstable system achieves stability**



Summary

BIBO Stability:

A system is called **Bounded-Input Bounded-Output (BIBO) stable**, or simply **stable**, if for any bounded input, its zero-state response is also bounded

Time-Domain Criterion:

$$\int_{-\infty}^{\infty} |h(t)| dt \leq M$$

(where M is a finite constant)

s-Domain Criterion: For $H(s) = L[h(t)]$, the system is stable if the region of convergence **includes the imaginary axis**

Causal and stable: **All poles of $H(s)$ lie in the left half of the s-plane**

1. Introduction

When introducing the Laplace transform, we addressed cases where $f(t)$ fails to satisfy the absolute integrability condition. By multiplying it with a decaying factor $e^{-\sigma t}$ and applying the Fourier transform, we derived the Laplace transform:

$$\mathcal{F}[f(t) \cdot e^{-\sigma t} u(t)] = \int_{0-}^{\infty} [f(t) \cdot e^{-\sigma t}] e^{-j\omega t} dt$$

$$s = \sigma + j\omega$$

$$\mathcal{L}[f(t)] = F(s)$$

2. Relationship Between FT and LT

(1) When $\sigma_0 > 0$, the convergence boundary lies in the **right** half of the s -plane

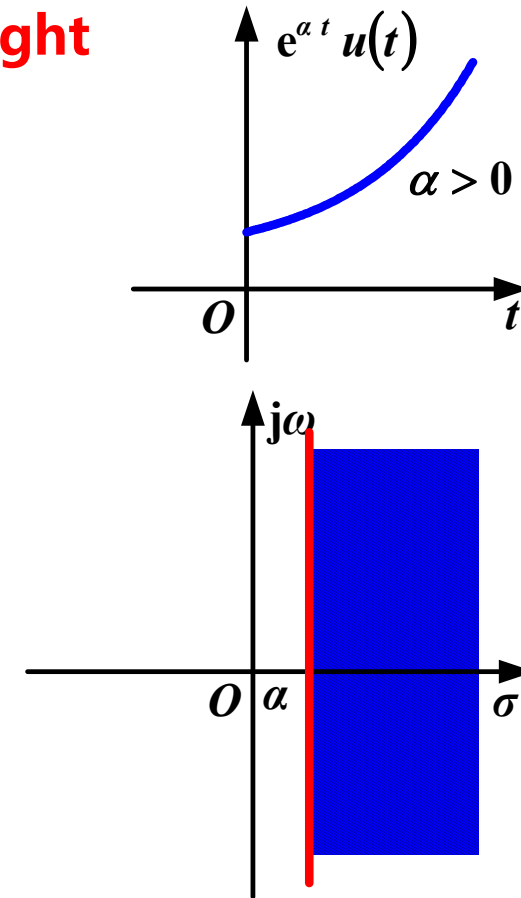
Consider the function:

$$f(t) = e^{\alpha t} u(t) (\alpha > 0)$$

Its Laplace transform is:

$$F(s) = \frac{1}{s - \alpha}$$

- Region of Convergence (ROC): $\sigma > \alpha$
- Fourier Transform Note:
 $F(j\omega)$ does **not exist** in this case
Thus, $F(j\omega)$ cannot be derived from $F(s)$



2. Relationship Between FT and LT

(2) When $\sigma_0 < 0$, the convergence boundary lies in the **left** half of the s -plane

Consider the decaying exponential function:

$$f(t) = e^{-\alpha t} u(t) (\alpha > 0)$$

Its Laplace transform is:

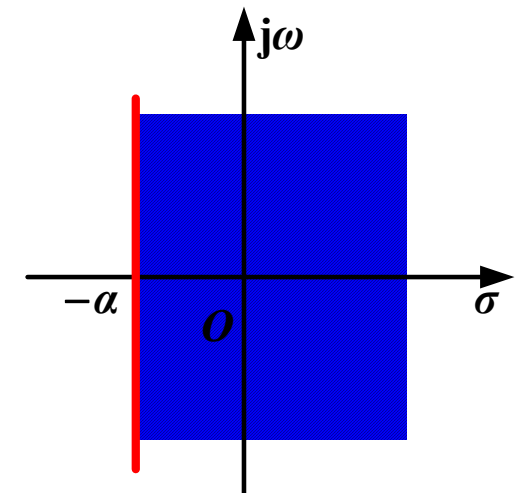
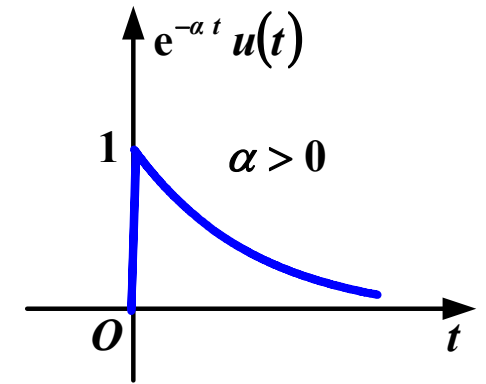
$$F(s) = \frac{1}{\alpha + s}$$

- Region of Convergence (ROC): $\sigma > -\alpha$

- Fourier Transform Existence:

Since $f(t)$ is a decaying function, its Fourier transform exists and can be obtained by evaluating $F(s)$ at $s = j\omega$:

$$F(j\omega) = F(s) |_{s=j\omega} = \frac{1}{\alpha + j\omega}$$

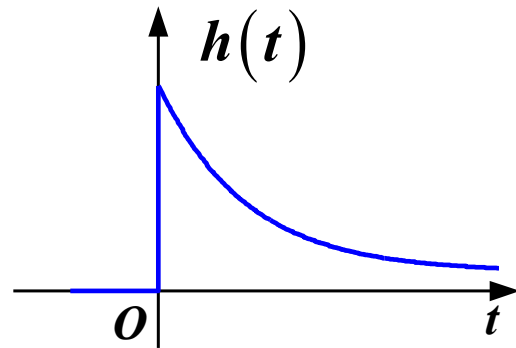


8 Relationship Between LT and FT

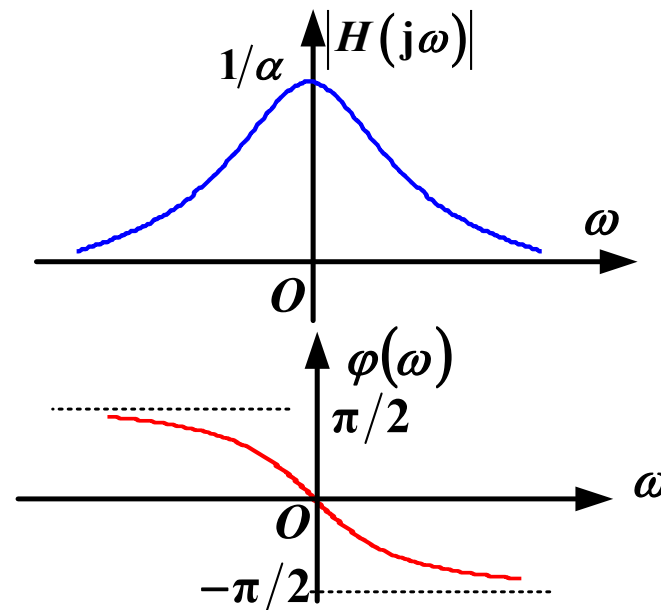
2. Relationship Between FT and LT

$$h(t) = e^{-\alpha t} \quad (\alpha > 0) \quad H(j\omega) = \frac{1}{j\omega + \alpha}$$

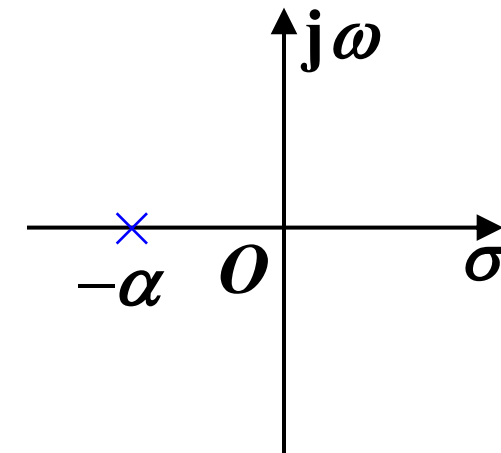
$$H(s) = \frac{1}{s + \alpha}$$



Time Domain



Frequency Domain



Pole-Zero Plot

2. Relationship Between FT and LT

(3) When $\sigma_0 = 0$, the convergence boundary lies on the **imaginary axis**

Key Observations: $F(s)$ exists, but $F(j\omega)$ cannot be obtained by simply substituting $s = j\omega$ in $F(s)$, as the Fourier transform includes singular function terms

Example

$$f(t) = u(t)$$

$$F(s) = \frac{1}{s}$$

$$F(j\omega) = \pi \delta(\omega) + \frac{1}{j\omega}$$

Summary

Relationship Between $F(s)$ and $F(j\omega)$:

Case 1: $\sigma_0 > 0$

Convergence axis lies in the right half of the s -plane, $F(j\omega)$ does not exist

Case 2: $\sigma_0 < 0$

Convergence axis lies in the left half of the s -plane, direct substitution is valid

$$F(j\omega) = F(s) |_{s=j\omega}$$

Case 3: $\sigma_0 = 0$

Convergence axis lies on the imaginary axis,

$$F(j\omega) = F(s) |_{s=j\omega} + \pi \sum_n k_n \delta(\omega - \omega_n)$$

Requires inclusion of impulse terms (Dirac delta functions)



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THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

Thank you for listening !

